

Qualcomm Technologies, Inc.

Device description

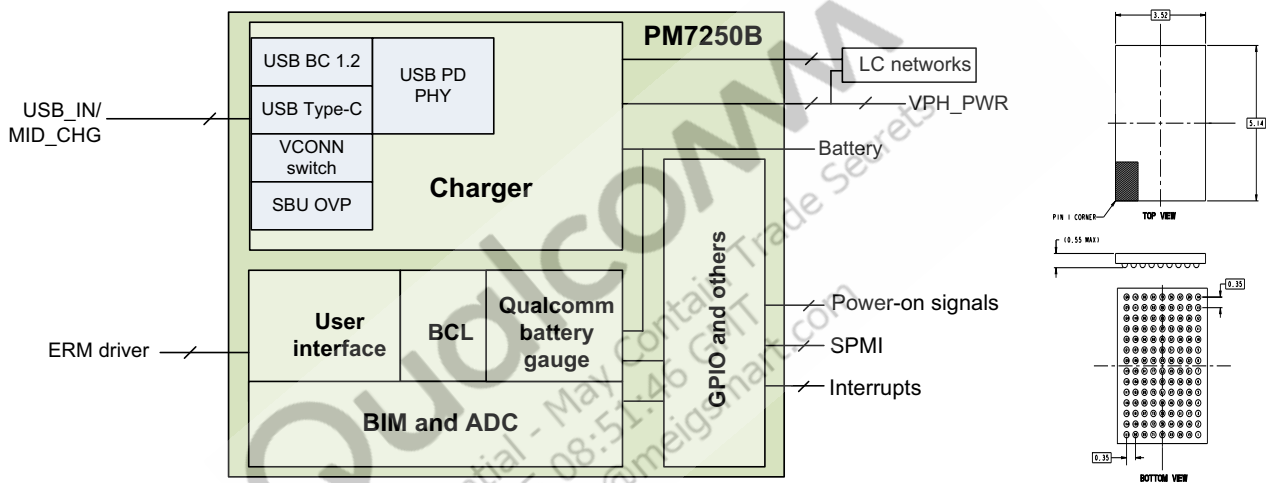
The PM7250B interface PMIC, along with its companion master core and slave core PMIC devices, integrates all wireless handset power management, general housekeeping, and user interface support functions.

- Best in class high-efficiency 4 A switch-mode Li-ion battery charger
- Supports 12 V input (28 V OVP) on USB
- Supports *USB Type-C Specification* Rev. 1.3 and *USB Power Delivery Specification* Rev. 3.0
- Supports Qualcomm® Quick Charge™ 2.0, Quick Charge 3.0, and Quick Charge 4.0 Technology

Key features (see Section 1.2 for details)

- Qualcomm® battery gauge with highly accurate battery state-of-charge estimation with aging and temperature correction. It is a hybrid of voltage and current based SoC and voltage loop feedback correction.
- Vibrator driver for ERM applications
- SPMI interface
- 12 GPIOs
- 110-pin FOWPSP

PM7250B high-level block diagram and FOWPSP110 package drawing



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1 Introduction

Document updates

See the [Revision history](#) for details on the changes included in this revision.

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1.1 Functional block diagram

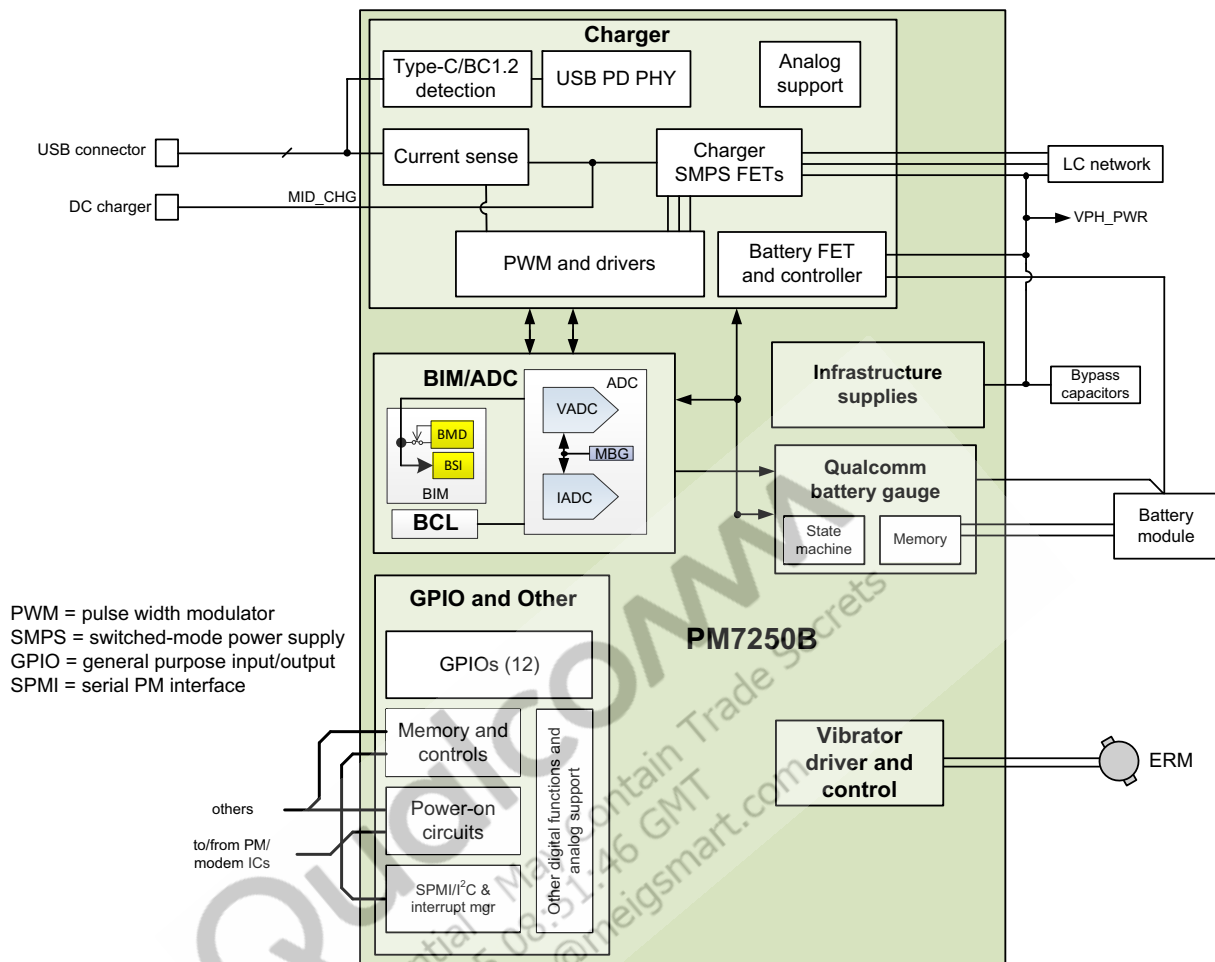


Figure 1-1 PM7250B high-level functional block diagram

1.2 PM7250B features

NOTE: Some hardware features integrated within the PM7250B device must be enabled through the IC software. See the latest revision of the applicable software release notes to identify the enabled PMIC features.

Table 1-1 PM7250B features

Feature	PM7250B capability
Charger	
Battery charger	■ High-efficiency 4 A switched-mode Li-ion battery charger □ Extremely low Rds(on) MOSFETs minimizing power dissipation during charging: – 12 mΩ USB FP FET, 17 mΩ HS FET, 19 mΩ LS FET, and 6 mΩ battery FET □ 600 kHz to 1.2 MHz programmable switching frequency allows optimizing tradeoff between power-train efficiency and external component selection, enabling 1.0 mm and 0.8 mm high BOMs □ 95.4% peak efficiency with 5 V input ■ USB charging input path □ 12 mΩ FP FET to minimize power loss during high current charging □ +3.6 V to +13.2 V operating voltage range on USB_IN with +28 V OVP □ USB Type-C CC logic compliant to the <i>USB Type-C Specification</i> Rev. 1.3. □ VCONN switch capable of delivering 1.5 W for DisplayPort alt-mode □ USB power delivery PHY hardware compliant to <i>USB Power Delivery Specification</i> Rev. 3.0 □ Quick Charge 2.0 and Quick Charge 3.0 hardware support; Quick Charge 4.0 software support □ Automatic power source detection (APSD) per the <i>USB Battery Charging Specification</i> Rev. 1.2 □ Automatic input current limit (AICL) for universal USB adapter compatibility ■ DC charging input path □ MID_CHG joint point, eliminating DC Front Porch FET or external PMUX □ Minimum MID_CHG voltage regulation for DC charging □ Automatic USB/DC input path selection and priority

Table 1-1 PM7250B features (cont.)

Feature	PM7250B capability
Battery charger (continued)	■ Interface to companion chargers □ Parallel charging support with SMB1390/SMB1355/SMB1396 ■ Battery interface □ 6 mΩ integrated BATFET for optimal charging efficiency and battery life □ Input/output CurrentPath control allows system operation with deeply discharged/missing battery □ Battery supplemental mode (BSM) to allow battery to supplement system current exceeding what the charger buck can provide □ Differential battery voltage Kelvin sensing to compensate for the IR drop caused by the phone PCB and battery pack protection FETs to prolong CC charging and reduce the total charging time □ JEITA battery temperature management and JISC 8714 support □ Battery-missing detection options (in the battery interface module) □ Ship-mode and AFP mode ■ Intelligent charging control and thermal management □ Hardware autonomous charging □ Trickle charger □ Hardware-controlled thermal loop manages input current limit based on the die temperature
Qualcomm battery gauge	
Qualcomm battery gauge	■ Highly accurate battery state-of-charge estimation with aging and temperature correction (approximately 2% of average state-of-charge error) ■ Simultaneous sampling for reading in-sync values of Vbatt and Ibatt with dedicated ADCs for battery voltage and current. ■ Complete battery cycling not required to maintain accuracy ■ Battery capacity learning and online equivalent series resistance (ESR) tracking
BIM and ADC	
BIM	■ Supports battery-missing detection ■ Three power modes: active, sleep, and off, providing optimized power consumption
On-chip ADC	Supports internal and external monitoring
Haptics	
Vibrator driver	■ Range of ERM parameters: R_ERM: 10 Ω to 30 Ω, L_ERM: 100 μH to 200 μH ■ Typical: 15 Ω and 150 μH ■ Vset range for this application: 1.504 V to 3.544 V ■ Supports only ERM, which is driven by LDO ■ Increased start up current limit to support high inrush from ERM ■ 500 mA current limit and 600 mA soft start current limit

Table 1-1 PM7250B features (cont.)

Feature	PM7250B capability
GPIO and other pins	
IC-level interfaces Primary status and control Interrupt managers Programmable boot sequence	- Two-line serial power management interface (MIPI SPMI) - Supported by SPMI - PBS 3.0 with one-time programmable (OTP) memory and user programmable - RAM for customizable power-on, power-off, and reset sequences
Configurable I/Os General-purpose input/output (GPIO) pins	12 GPIO pins, configurable as digital inputs or outputs - Some GPIOs have primary/alternate functions for IC-level Interfacing or required controls for user interface functions if those UI functions are used.
Internal clocks	Internal 19.2 MHz RC clock
Package	
Size	3.51 × 4.40 × 0.55 mm
Pin count and package type	FOWPSP110 (0.35 mm pitch)

1.3 PM7250B SKUs

PM7250B has the following SKUs: PM7250B and PM7250B-2.

Features wise there are no differences between both the SKUs, they are used independently in different chipsets.

Table 1-2 PM7250B SKUs

	Variant
Chipset attaches	Below programs can use PM7250B-0 or PM7250B-2 SKU: - SM4375 - SM6375 - SM7250 - SM7225 and SM6350 - SM4350 Except the programs mentioned above, all other programs must use PM7250B-2.

NOTE: Do not use any SKU chip in a different chipset platform other than what is recommended by QTI or mentioned in the respective reference schematic.

2 Pad definitions

The PM7250B is available in the FOWPSP110 – see [Chapter 4](#) for package details. A high-level view of the pad assignments is shown in [Figure 2-1](#).

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Figure 2-1 PM7250B pad assignments (top view)

2.1 I/O parameter definitions

Table 2-1 I/O description (pad type) parameters

Symbol	Description
Pad attribute	
AI	Analog input
AO	Analog output
DI	Digital input (CMOS)
DO	Digital output (CMOS)
PI	Power input; a pad that handles 10 mA or more of current flow into the device ¹
PO	Power output; a pad that handles 10 mA or more of current flow out of the device ¹
Z	High-impedance (Hi-Z) output
MV	Medium voltage
LV	Low voltage
GNDP	Power ground; a pad that handles 10 mA or more of current flow returning to ground. Layout considerations must be made for these pads.
GNDG	Common ground; a pad that does not handle a significant amount of current flow, typically used for grounding digital circuits and substrates.
GPIO pins, when configured as outputs, have configurable drive strengths that depend upon the GPIO pad's supply voltage. See the electrical specifications in Chapter 3 for details.	

1. The maximum current levels expected on PI and PO type pads are listed in [Chapter 3](#).

2.2 Pad descriptions

Descriptions of all pins are presented in the following tables, organized by functional group:

Table 2-2 [Pad descriptions – input power management functions](#)

Table 2-3 [Pad descriptions – General HK functions](#)

Table 2-4 [Pad descriptions – General purpose input/output functions](#)

Table 2-5 [Pad descriptions – IC-level interface functions](#)

Table 2-6 [Pad descriptions – User interface functions](#)

Table 2-7 [Pad descriptions – common grounds](#)

Table 2-2 Pad descriptions – input power management functions

Pad #	Pad name	Pad type ¹	Functional description
Charger/OTG interface			
1, 9, 18, 26, 35, 43	USB_IN	PI, PO	Input power from the selected source (USB or DC_IN), or output during USB-OTG. This is a power entry node for the charger and connects to the OVP circuitry.
60	IUSB_OUT	AO	Buffered voltage signal proportional to USB input current. Output to the charge pump. Needed for regulating the input current limit with the charge pump.
73	USB_DP	AI, AO	USB data plus for power source detection only; the modem IC handles data transactions.
81	USB_DM	AI, AO	USB data minus for power source detection only; the modem IC handles data transactions.
Switching charger (SCHG)			
2, 10, 19, 27, 36, 44, 52	MID_CHG	PI	Mid point of the charger. Charger buck input. Joint point for DC/charging.
3, 11, 20, 28, 29, 37	VSW_CHG	PO	Charger SMPS switching node.
4, 12, 21, 38	PGND_CHG	GNDP	Specific ground for the SCHG. Layout considerations must be made for this pad.
5, 13, 22, 30, 39, 56	VPH_PWR	PI, PO	Primary system supply node, SCHG regulated node
6, 14, 23, 31, 40, 48	VBATT_PWR	PI, PO	Battery voltage node, connects to BATFET. Output is for charging, and input is for all other operations.
25	ISNS_SMB_M	AI	Current sense minus from the external parallel charger.
34	ISNS_SMB_P	AI	Current sense plus from the external parallel charger.
32	PACK_SNS_M	AI	Battery voltage sense input minus. Reference point for VADC for measuring battery voltage (VBATT_SNS_P). Connect this directly to the battery negative node (pack negative).
45	BOOT_CAP	AO	Charger bootstrap node for bootstrapping the charger startup bias network with input power before starting the SCHG
53	BOOT_PWR	AO	Auxiliary 5 V to 6 V low dropout (LDO) output – 50 mA (minimum) output supply for the boot strap capacitor.
55	SBU1	DI	Type-C side band signal SBU1; protected to 22 V max.
46	SBU2	DO	Type-C side band signal SBU2; protected to 22 V max.
61	SMB_EN	DO	Enable/disable control pin for parallel SMB (slave) charger.
62	VDD_VCONN	PI	VCONN input source, externally connected to VPH, VBOB, or dedicated boost converter.
47	VBATT_SNS_M	AI	Battery voltage sense input minus. Reference point used by the charger for battery voltage sensing. Connect to the battery negative remote sense node or connect this directly to the battery negative node.
64	VBATT_SNS_P	AI	Battery voltage sense input plus. Connect to the battery positive remote sense node or connect this directly to the battery positive node.
54	CC1_ID	AI, PO	OTG mode enable or CC1 pin for the USB Type-C connector (user programmable). Requires IEC protection.

Table 2-2 Pad descriptions – input power management functions (cont.)

Pad #	Pad name	Pad type ¹	Functional description
63	CC2	AI, PO	CC2 pin for the USB Type-C connector. Requires IEC protection.
72	CC_OUT	DO	1.8 V push-pull tri-state output indicating CC1 or CC2 connection (orientation).
88	VDD_PDPHY	PI	Power supply to USB PD PHY.
96	REF_GND_CHG	GNDP	Dedicated ground for the charger-specific master bandgap. Special considerations must be made to ensure that this ground is properly connected on the PCB.
100	PVDD	PI	Connects to VPH_PWR. Powers internal PMIC circuits.
DC_IN charging			
69	DC_IN_PSNS	AI	DC charging power sense input.
77, 94	DC_IN_PON	AI	DC charging power-on trigger
86, 103	DC_IN_EN	DO	DC barrel jack charging power source enable/disable pin. Keeps DC charger input disabled during USB charging.
Qualcomm battery gauge/battery interface			
84	BA_N	DO	Battery alarm open drain output. Part of the (BCL) system.
7	BATT_ID	AI	Battery ID input to the ADC interface. It can be used for missing battery detection.
15	BATT_THERM	AI	Battery temperature input to ADC for measuring the pack temperature. It is used for charger safe operation and BMS.
24	SMB_THERM	AI	Inductor temperature sensor.
33	CONN_THERM	AI	USB Type-C connector temperature sensor.
79	VARB_CHG	PO	Power source arbitration circuit output.

1. See Table 2-1 for parameter and acronym definitions.

Table 2-3 Pad descriptions – General HK functions

Pad #	Pad name	Pad type ¹	Functional description
Analog multiplexer and HK/XO ADC circuits			
8, 16, 17	GND_ADC	GNDP	Ground for ADC circuits. Needs layout considerations to keep it quiet.
49	AMUX_1	AI	Analog multiplexer (AMUX) input
42	OPT_M	AI	Reserved.
51	OPT_P	AI	Reserved.
PMIC infrastructure circuits			
50	AVDD_BYP	AO	Bypass capacitor connection for the internal aVdd regulator (5 V); used to power internal analog infrastructures.
58	DVDD_BYP	AO	Bypass capacitor connection for the internal dVdd regulator (1.5 V); used to power internal digital infrastructures.

Table 2-3 Pad descriptions – General HK functions (cont.)

Pad #	Pad name	Pad type ¹	Functional description
59	REF_BYP	AO	Bypass capacitor for the dedicated master bandgap regulator; this reference must only be used for the master bandgap and must not be used as a general reference output.
67	REF_GND	GNDP	Dedicated master bandgap ground reference connection; should be isolated from all other grounds – the pad should be connected directly to the main ground plane (with a via at the pin) and to the REF_BYP capacitor negative terminal.

1. See Table 2-1 for the parameter and acronym definitions.

Table 2-4 Pad descriptions – General purpose input/output functions

Pad #	Pad name	Configurable function	Pad type ¹	Functional description
Configurable GPIOs				
106	GPIO_1	Spare	LV	Configurable; default digital input with 10 μ A pull-down
104	GPIO_2	Spare	LV	Configurable; default digital input with 10 μ A pull-down
87	GPIO_3	Spare	LV	Configurable; default digital input with 10 μ A pull-down
97	GPIO_4	Spare	LV	Configurable; default digital input with 10 μ A pull-down
66	GPIO_5	Spare	MV	Configurable; default digital input with 10 μ A pull-down
98	GPIO_6	SMB_STAT	MV	Configurable; default digital input with 10 μ A pull-down SMB1355/SMB1390/SMB1396 status
99	GPIO_7	Spare	MV	Configurable; default digital input with 10 μ A pull-down
68	GPIO_8	Spare	MV	Configurable; default digital input with 10 μ A pull-down
95	GPIO_9	FMB1	LV	Configurable; default digital output, open drain PMOS Factory mode boot 1
105	GPIO_10	FMB2	LV	Configurable; default digital output, open drain NMOS Factory mode boot 2
75	GPIO_11	Spare	LV	Configurable; default digital input with 10 μ A pull-down
107	GPIO_12	Spare	LV	Configurable; default digital input with 10 μ A pull-down

1. See Table 2-1 for the parameter and acronym definitions.

Table 2-5 Pad descriptions – IC-level interface functions

Pad #	Pad name	Pad type ¹	Functional description
Power-on/off/reset			
41	OPTION	AI	Option pin is used to select different PON options based on pull-down (PD) resistor value
65	SYS_OK	DO	Upon charger insertion, the charger system sets SYS_OK high and the PON module triggers the PBS PON sequence.
78	KPD_PWR_N	DI	Exits ship mode when pressed; does not trigger power on

Table 2-5 Pad descriptions – IC-level interface functions (cont.)

Pad #	Pad name	Pad type ¹	Functional description
108	FAULT_N	DI/DO	PMIC fault signal (bidirectional) that initiates shutdown or S3 reset to all PMICs.
System power management interface (SPMI) signals			
93, 102	SPMI_CLK	DI	SPMI communication bus clock signal
101, 110	SPMI_DATA	DI, DO	SPMI communication bus data signal
Digital I/O input supply			
92	TEST_EN_VPP	GNDC	Pin must be grounded externally
109	VDD_IO	PI	1.8 V input supply power for digital I/Os

1. See [Table 2-1](#) for the parameter and acronym definitions.

Table 2-6 Pad descriptions – User interface functions

Pad #	Pad name	Pad type ¹	Functional description
Haptics			
76	VIB_DRV_P	PO	Power supply for haptics driver
85	VDD_VIB_DRV	PI	Input for VIB_DRV

1. See [Table 2-1](#) for the parameter and acronym definitions.

Table 2-7 Pad descriptions – common grounds

Pad #	Pad name	Pad type ¹	Functional description
Common grounds			
71	GND_CHG	GNDC	SMBC controller ground
57, 74, 80, 82, 83, 89, 90, 91	CMN_GND	GNDC	Ground for all non-specialized circuits
70	GND_PSUB_CHG	GNDC	Substrate ground for the SMBC

1. See [Table 2-1](#) for the parameter and acronym definitions.

3 Electrical specifications

3.1 Absolute maximum ratings

The absolute maximum ratings (Table 3-1) reflect the stress levels that, if exceeded, may cause permanent damage to the device. No functionality is guaranteed outside the operating specifications. Functionality and reliability are only guaranteed within the operating conditions described in Section 3.2.

Table 3-1 Absolute maximum ratings

Parameter	Min	Max	Units
Terminal voltage (reference to GND unless otherwise noted)			
USB_IN, MID_CHG ¹	-0.3	28	V
DC_IN_PON ¹	-0.3	16	V
MID_CHG (reference to USB_IN) (that is, FP FET V _{DS})	-0.3	16	V
VSW_CHG ¹	-0.3	22	V
CC1, CC2, SBU1, SBU2 ¹	-0.3	22	V
VPH_PWR, VBATT_PWR	-0.3	6	V
VPH_PWR, VBATT_PWR (10 ms transient)	-0.3	7	V
All others	-0.3	Lower of +6 V and (V _{xx} + 0.5 V) ²	V
Storage temperature (T _S) ³	-55	+150	°C

1. Battery voltage of at least 2.5 V is present on VBATT_PWR

2. V_{xx} is USB_IN, DCIN_PON or VBATT_PWR

3. The storage temperature range applies when the device is in the OFF state (the device is not assembled in any platform and is not electrically connected to any voltage or I/O signals). Damage may occur when the device is subjected to this temperature for any length of time.

NOTE: The device needs to be powered/biased via USB_IN, DCIN_PON, or VBATT_PWR before applying power on any of the other output I/Os.

3.2 Operating conditions

Operating conditions include design team-controlled parameters such as power supply voltage and thermal conditions (Table 3-2). The PM7250B meets all performance specifications listed in Section 3.3 through Section 3.11, when used within the operating conditions, unless otherwise noted in those sections (provided the absolute maximum ratings have never been exceeded).

Table 3-2 Operating conditions

Parameter		Min	Typ	Max	Units
Thermal conditions					
T _C					
	Case operating temperature	-30	+25	+85	°C
Input power pads					
USB_IN	Input power from USB source	3.7	–	12.6	V
MID_CHG	MID point of the charger. Charger buck input. Joint point for DC/wireless charging and SMB parallel charger(s).	3.7	–	12.6	V
DC_IN_PON	DC/wireless charging power-on trigger	3.7	–	12.6	V
VPH_PWR	Handset power-supply voltage	2.7	3.8	4.8	V
VBATT_PWR VBATT_SNS_x	Main battery voltage	2.7	3.8	4.8	V

3.3 DC power consumption

This section specifies DC power supply currents for the various IC operating modes (Table 3-3). Typical currents are based on IC operation at room temperature (+25°C) using default settings.

Table 3-3 DC power supply currents

Parameter		Min	Typ	Max	Units
IDD _{ACTIVE}	Battery supply current, active mode ¹	–	280	–	μA
IDD _{SLEEP}	Battery supply current, sleep mode ²	–	280	–	μA
IDD _{OFF}	Battery supply current, off mode ³	–	27.5	–	μA
IDD _{SHIP}	Battery supply current, ship mode ⁴	–	37	–	μA

1. IDD_{ACTIVE} is the total supply current from the main battery with PM7250B on, the Qualcomm battery gauge in an active state, charger in standby, and haptics modules are off. SBL has been loaded.
2. IDD_{SLEEP} is the total supply current from a main battery with PM7250B on, the Qualcomm battery gauge in low power mode, the charger in standby, sleep_b = 1, haptics modules are off. SBL is loaded, and the MPM sleep sequence is executed. BAT_ID pulldown = 100 kΩ, BAT_THERM = 68.1 kΩ, and VBAT = 3.7 V. This only applies when the temperature is between -30°C and 60°C.
3. IDD_{OFF} is the total supply current from the main battery with the PM7250B off. This only applies when the temperature is between -30°C and +60°C.
4. IDD_{SHIP} is the total supply current from a main battery with PM7250B off and in ship mode. BATFET is open. This only applies when the temperature is between -30°C and 60°C.

3.4 Digital logic characteristics

The charger has unique digital signaling characteristics as listed within [Section 3.5](#). All other PM7250B digital I/O characteristics are specified in [Table 3-4](#).

Table 3-4 Digital I/O specifications

Parameter		Condition	Min	Max	Units
V_{IH}	High-level input voltage ¹	–	$0.65 \times V_{IO}$	$V_{IO} + 0.3$	V
V_{IL}	Low-level input voltage ¹	–	-0.3	$0.35 \times V_{IO}$	V
V_{OH}	High-level output voltage ¹	$I_{OUT} = I_{OH}$	$V_{IO} - 0.45$	V_{IO}	V
V_{OL}	Low-level output voltage	$I_{OUT} = I_{OL}$	0	0.45	V
V_{HYST}	Schmitt trigger hysteresis	–	15	–	mV
I_{OH}	High-level output current ²	–	3	–	mA
I_{OL}	Low-level output current ²	–	–	-3	mA
I_L	Input leakage current, digital input ³	–	-200	200	nA
C_{IN-D}	Input capacitance, digital input or input/output	–	–	5	pF

1. $V_{IO} = VDD_{IO}$, unless otherwise noted.
2. The output current specifications apply to all digital outputs, unless specified otherwise, and are superseded by specifications for specific pins.
3. GPIO pins comply with the input leakage specification only when configured as a digital input or set to the tri-state mode.

3.5 Charger

The PM7250B features a fully programmable switched-mode Li-ion battery charger and input power and output power controller for portable devices. The device is designed to be used with systems using single-cell Li-ion and Li-polymer battery packs. The PM7250B provides three major functions to the end-system: input selection and arbitration, system output supply and control, and battery charging. The device is fully programmable via the SPMI interface.

3.5.1 Input power specifications

Table 3-5 Input power source control and protection

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
$V_{\text{COARSE_DET}}$	USB/DC input coarse detection threshold	$V_{\text{USB_IN}}/V_{\text{DCIN PON}}$ rising and falling	0.7	1.0	1.3	V
	Deglintch filter time	$V_{\text{USB_IN}}/V_{\text{DCIN PON}}$ rising and falling	–	10	–	μS
V_{UVLO}	USB/DC input undervoltage lockout (UVLO) threshold	V_{IN}^1 falling; for 5 V options	3.5	3.6	3.7	V
		V_{IN}^1 falling; for 9 V options	6.6	6.9	7.2	V
		V_{IN}^1 falling; for 12 V options	9.3	9.6	9.9	V
	Hysteresis	V_{IN}^1 rising	–	200	–	mV
	Deglintch filter time	V_{IN}^1 rising and falling option 1	–	10	–	ms
		V_{IN}^1 rising and falling option 2	–	20	–	ms

Table 3-5 Input power source control and protection (cont.)

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
V _{OVLO}	USB/DC input overvoltage lockout (OVLO) threshold	V _{IN} ¹ rising; for 5 V only (option 1)	6.2	6.4	6.6	V
		V _{IN} ¹ rising; for 5 V only (option 2)	6.9	7.2	7.5	V
		V _{IN} ¹ rising; for 9 V only or 5 V to 9 V	10.5	10.8	11.1	V
		V _{IN} ¹ rising; for 9 V to 12 V, 12 V only, continuous (option 1)	12.6	13.2	14	V
		V _{IN} ¹ rising; for 9 V to 12 V, 12 V only, continuous (option 1)	14	14.6	15.2	V
	Hysteresis	V _{IN} ¹ falling	–	200	–	mV
	Response time	V _{IN} ¹ rising at 5 V/μs, 12 V range	–	–	2	μs
		V _{IN} ¹ rising at 5 V/μs, all ranges except 12 V	–	10	–	μs
	Deglitch filter time	V _{IN} ¹ falling	–	100	–	ms
V _{ASHDN}	USB/DC input auto-shutdown threshold	V _{IN} ¹ - V _{SYS} , V _{IN} ¹ falling	70	130	190	mV
	Hysteresis	V _{IN} ¹ rising	–	80	–	mV
	Deglitch filter time	V _{IN} ¹ rising and falling	–	20	–	ms
V _{REVI}	USB/DC input reverse-current threshold	V _{IN} ¹ - V _{SYS} , V _{IN} ¹ falling	-70	-130	-190	mV
	Hysteresis	V _{IN} ¹ rising	–	80	–	mV
	Deglitch filter time	V _{IN} ¹ falling	–	100	–	μs
		V _{IN} ¹ rising	–	1	–	ms
I _{PD_USB_IN}	USB_IN pull-down current source	V _{USB_IN} OV or UV	5	10	15	mA
I _{PD_MID_CHG}	MID_CHG pull-down current source	V _{USB_IN} UV	5	10	15	mA
I _{PD_DCIN_PON}	DCIN_PON pull-down current source		5	10	15	mA

Table 3-5 Input power source control and protection (cont.)

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
R _{PD_DCIN_PON}	DCIN_PON pull-down resistor		7	10	13	kΩ
	USB_IN surge protection	IEC 61000-4-5 1.2/50 μs voltage surge waveform, external TVS diode	–	–	400	V
	Maximum FP FET VDS during surge	IEC 61000-4-5 1.2/50 μs voltage surge waveform, external TVS diode	–	–	16	V
I _{SUSP_USB}	USB_IN suspend mode current	USB_IN path in suspend mode, V _{USB_IN} = 5 V, V _{BAT} = 4.5 V	–	1.4	2.2	mA
I _{SUSP_DC}	DC suspend mode current	DC path in suspend mode, V _{UVLO} < V _{MID_CHG} < V _{OVLO}	–	2.0	2.8	mA
I _{LIM_USB}	USB path maximum input current limit	USB 2.0 option: 100 mA mode, T = 0°C to +70°C, USB_IN = 5 V	70	85	100	mA
		USB 3.0 option: 150 mA mode, T = 0°C to +70°C, USB_IN = 5 V	115	–	150	mA
		USB 2.0 option: 500 mA mode, T = 0°C to +70°C, USB_IN = 5 V	450	475	500	mA
		USB 3.0 option: 900 mA mode, T = 0°C to +70°C, USB_IN = 5 V	810	855	900	mA
		Type-C 1.5 A mode, T = 0°C to +70°C, USB_IN = 5 V	1350	1425	1500	mA
		Type-C 3.0 A mode, T = 0°C to +70°C, USB_IN = 5 V	2700	2850	3000	mA
		USB high-current mode (0 mA to 5.0 A programmable, 50 mA steps), USB_IN = 5 V, 0.5 A ≤ I _{LIM_USB} ≤ 5 A, T = 0°C to +70°C	0.9 * I _{LIM_USB}	0.95 * I _{LIM_USB}	I _{LIM_USB}	mA
		USB high current mode, USB_IN = 9 V, 1 A ≤ I _{LIM_USB} ≤ 3 A, T = 0°C to +70°C	0.87 * I _{LIM_USB}	0.935 * I _{LIM_USB}	I _{LIM_USB}	mA
		USB high current mode, USB_IN = 12 V, 1 A ≤ I _{LIM_USB} ≤ 1.5 A, T = 0°C to +70°C	0.87 * I _{LIM_USB}	0.935 * I _{LIM_USB}	I _{LIM_USB}	mA

Table 3-5 Input power source control and protection (cont.)

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
V _{IUSB_OUT}	IUSB_OUT pin V/A ratio	Buck mode, ICL > 100/150 mA	–	0.2	–	V/A
		Buck mode, ICL = 100/150 mA	–	1	–	V/A
		Boost mode	–	0.4	–	V/A
	Accuracy	I _{USB} ≥ 500 mA, T = 0°C to +70°C	0	–	10	%
DC/wireless input						
V _{PSNS_MAX}	DC path maximum input power limit on DC_IN_PSNS	Programmable range (30 settings in 40 mV steps); 15 W/1.2 V	0	–	1.2	V
	Accuracy	P _{MAX_DC} ≥ 5 W, T = 0°C to +70°C; actual-to-setting ratio	98	100	102	%
V _{PSNS_THR}	PSNS threshold voltage	V _{PSNS} < V _{PSNS_THR} , PSNS regulation; V _{PSNS} > V _{PSNS_THR} , VMID regulation	1.65	–	1.7	V
V _{MID_MIN}	DC path minimum MID_CHG voltage regulation	Programmable range (64 settings in 100 mV steps)	4	–	10.3	V
	Accuracy	T = 0°C to +70°C	-2	–	2	%
Automatic input current limiting (AICL)						
V _{AICL}	5 V AICL voltage collapse threshold	Programmable range (8 settings in 100 mV steps); V _{USB_IN} falling	4	–	4.7	V
	9 V AICL voltage collapse threshold	Programmable range (8 settings in 200 mV steps); V _{USB_IN} falling	7.2	–	8.6	V
	12 V AICL voltage collapse threshold	Programmable range (8 settings in 200 mV steps); V _{USB_IN} falling	10.2	–	11.6	V
	AICL voltage collapse threshold - continuous mode	Programmable range (100 mV steps between 4.0 V to 5.6 V; 200 mV steps between 5.8 V to 11.8 V); V _{USB_IN} falling	4	–	11.8	V
	Accuracy	V _{USB_IN} falling	-3.5	–	3.5	%
	Hysteresis	V _{USB_IN} rising	–	200	–	mV
	Deglint filter time	V _{USB_IN} rising and falling; Programmable in 30 μs, 1, 5, 30 ms	0.03	5	30	ms

Table 3-5 Input power source control and protection (cont.)

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
T_{AICL_step}	AICL step duration in discrete method	USB_IN ICL rising; Programmable in 5, 10, 30, 40 ms	5	10	40	ms
T_{AICL_rerun}	AICL rerun timer	Programmable in 3, 12, 45, 360 sec	3	45	360	S

1. For the PM7250B charger, VIN is VUSB_IN for the USB path or VMID_CHG for the DC path (no DC FP FET).

Table 3-6 USB Type-C and BC1.2 specifications

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
Automatic power source detection (APSD) for BC 1.2						
V _{DP_SRC}	D+ source voltage	I _{DP_SRC} < 250 μA	0.6	0.65	0.7	V
V _{DAT_REF}	Data detect voltage		0.3	0.35	0.4	V
V _{DP_UP}	D+ pull-up voltage		3	3.3	3.6	V
I _{DM_SINK}	D- sink current		25	50	75	μA
I _{DP_SRC}	Data contact detect current source		7	–	10	μA
t _{DP_SRC_ON}	D+ source on time		100	–	–	ms
t _{DCD_TIMEOUT}	DCD timeout	Option 1	321	328	335	ms
		Option 2	642	656	670	ms
t _{ENUM_TIMEOUT}	SDP enumeration timeout		–	–	2	min
t _{DPSRC_HICRNT}	D+ source off to high current		40	–	–	ms
t _{DPSRC_CON}	D+ source off to connect		40	–	–	ms
C _{DP/DM}	D+/D- capacitance	D+/D- Hi-Z	–	4	–	pF
R _{ID_GND}	ID resistance ground detection	Micro USB mode	–	–	1.3	kΩ
R _{ID_FMB_1}	Micro USB factory mode boot resistance range 1	V _{USB_IN} present, micro USB mode	1.8	–	178	kΩ
R _{ID_FMB_2}	Micro USB factory mode boot resistance range 2	V _{USB_IN} present, micro USB mode	274	–	665	kΩ
R _{ID_FLOAT}	ID resistance float detection	V _{USB_IN} present, micro USB mode	210	–	–	kΩ
		V _{USB_IN} absent, only battery present, micro USB mode	215	–	226	kΩ
		V _{USB_IN} absent, only battery present, micro USB mode	806	–	–	kΩ
USB Type-C interface and detection						
CC _{CAP}	Maximum CC pin capacitance		200	–	600	pF
Z _{OPEN}	Sink minimum open-circuit CC impedance to ground	In Disabled or ErrorRecovery state	126	–	–	kΩ

Table 3-6 USB Type-C and BC1.2 specifications (cont.)

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
$V_{CC_CLAMP_SNK}$	Sink CC clamp voltage	As Rd during shutdown	0.88	1.1	1.32	V
Rd	Sink Rd pull-down resistor		4.59	5.1	5.61	k Ω
I_{RP_SRC}	Source Rp current source – default USB		64	80	96	μ A
	Source Rp current source – 1.5 A		166	180	194	μ A
	Source Rp current source – 3.0 A		304	330	356	μ A
vSafe0V			0	–	0.8	V
vSafe5V			4.75	–	5.5	V
$V_{BUS_RMV_5V}$	Sink VBUS removal detection threshold if = 5 V (vSinkDisconnect)	V_{USB_IN} falling; for determining exit from Attached.SNK state	0.8	2.5	3.67	V
	Hysteresis	V_{USB_IN} rising	–	500	–	mV
$V_{BUS_RMV_9V}$	Sink VBUS removal detection threshold if = 9 V	V_{USB_IN} falling; for determining exit from Attached.SNK state	6.6	6.9	7.2	V
	Hysteresis	V_{USB_IN} rising	–	200	–	mV
$V_{BUS_RMV_12V}$	Sink VBUS removal detection threshold if = 12 V	V_{USB_IN} falling; for determining exit from Attached.SNK state	9.3	9.6	9.9	V
	Hysteresis	V_{USB_IN} rising	–	200	–	mV
Vconn switch						
V_{CONN}	V_{CONN} voltage range	V_{CONN} enabled, $3.2\text{ V} \leq V_{PH_PWR} \leq 4.5\text{ V}$	3.0	–	4.5	V
	V_{CONN} load	V_{CONN} enabled, $V_{PH_PWR} \geq 3.2\text{ V}$	1.5	–	–	W
I_{OCP_VCONN}	V_{CONN} overcurrent protection threshold	V_{CONN} enabled, $V_{PH_PWR} \geq 3.2\text{ V}$ T = 0°C to +70°C, 100mA option	100	175	250	mA
		V_{CONN} enabled, $V_{PH_PWR} \geq 3.2\text{ V}$ T = 0°C to +70°C, 500 mA option	500	625	750	mA
$V_{DIP_VCONN_SS}$	V_{PH_PWR} dip when V_{CONN} soft-start	10 μ F V_{CONN} load; enable V_{CONN}	0	–	200	mV
SBU						
I_{SBU_SRC1}	SBU current source 1	For Type-C connector water detection	8	10	12	μ A
I_{SBU_SRC2}	SBU current source 2	For Type-C connector water detection	160	180	200	μ A

3.5.2 Battery charger specifications

Table 3-7 Battery charger specifications

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
Battery charger						
V_{TRKL}	Trickle charge to pre-charge threshold	–	2	2.1	2.2	V
I_{TRKL}	Trickle charge current	$V_{BATT} < V_{TRKL}$	35	50	65	mA
I_{PRE_CHG}	Pre-charge current range	Programmable (8 settings in 50 mA steps)	100	300	450	mA
	Accuracy	$T = 0^{\circ}\text{C to } +70^{\circ}\text{C}$ $I_{pre_chg} < 250 \text{ mA}$	$\pm 15\% + 20 \text{ mA}$			
		$T = 0^{\circ}\text{C to } +70^{\circ}\text{C}$ $I_{pre_chg} \geq 250 \text{ mA}$	$\pm 15\% + 15 \text{ mA}$			
I_{FC}	Fast charge current (FCC) range ¹	Programmable (160 settings in 50 mA steps)	0	–	8 ²	A
	Accuracy	$T = 0^{\circ}\text{C to } +70^{\circ}\text{C}$, $I_{FC} < 1.0 \text{ A}$	-12.5	–	12.5	%
		$T = 0^{\circ}\text{C to } +70^{\circ}\text{C}$, $1.0 \text{ A} \leq I_{FC} < 2.0 \text{ A}$	-7.5	–	7.5	%
		$T = 0^{\circ}\text{C to } +70^{\circ}\text{C}$, $I_{FC} > 2.0 \text{ A}$	-5	–	5	%
		$T = 25^{\circ}\text{C}$, $I_{FCH} < 1.0 \text{ A}$	-10	–	10	%
		$T = 25^{\circ}\text{C}$ $1.0 \text{ A} \leq I_{FCH} < 2.0 \text{ A}$	-5	–	5	%
		$T = 25^{\circ}\text{C}$, $I_{FCH} > 2.0 \text{ A}$	-5	–	5	%
V_{FLT}	Float voltage range	Programmable (120 settings with 10 mV steps), $T=0^{\circ}\text{C to } +70^{\circ}\text{C}$	4.0	–	4.55	V
	Accuracy	$V_{FLT} < 4.2 \text{ V}$, $T = 0^{\circ}\text{C to } +70^{\circ}\text{C}$	-1	–	1	%
		$V_{FLT} \geq 4.2 \text{ V}$, $T = 0^{\circ}\text{C to } +70^{\circ}\text{C}$	-0.5	–	0.5	%
R_{AFVC}	AFVC ESR	Programmable (in 2 m Ω steps)	0	20	62	m Ω
V_{AFVC}	AFVC compensation voltage	Accuracy (including both I_{CHG} and R_{AFVC} inaccuracies)	-10	–	10	%

Table 3-7 Battery charger specifications (cont.)

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
I _{TERM}	Charge termination current range	IADC range	50	–	750	mA
	Accuracy	IADC internal current sense, VBATT = 4.2 V-4.55 V, I _{TERM} = 100 mA, T = 0°C to +70°C	75	100	125	mA
I _{TERM_ANA}	I _{TERM} analog comparator range	Programmable (8 settings in 50 mA steps)	100	300	450	mA
	Accuracy	T = 0°C to +70°C	± 10% of setting ± 50 mA			
V _{RECHG}	Voltage-based automatic recharge threshold	Programmable VBAT range	3.6	–	4.8	V
	Accuracy	T = 0°C to +70°C	-20	–	20	mV
SoC _{RECHG}	SoC-based automatic recharge threshold	Programmable SoC range	0	–	100	%
	Accuracy	T = 0°C to +70°C	-1	–	1	%
V _{INHIBIT}	Charge inhibit threshold voltage range	Programmable (50 mV, 100 mV, 200 mV, and 300 mV)	50	–	300	mV
	Accuracy	T = 0°C to +70°C	150	–	200	mV
V _{LOW_BATT}	Low-battery voltage/SYSOK detection (triggers PMIC power-on) threshold range	Programmable (16 settings in 50 mV steps), VBATT falling	2.5	2.8	3.25	V
	Accuracy	V _{LOWBATT} = 2.8 V	-2	–	2	%
	Hysteresis	V _{BATT} rising	–	200	–	mV
V _{BATT_OV}	Battery overvoltage lockout	V _{BATT} rising, above V _{FLT} + AFVC	70	100	130	mV
	Hysteresis	V _{BATT} falling	35	50	65	mV
V _{SYS_HV_ALM}	V _{SYS} high-voltage alarm	Programmable (8 settings in 100 mV steps)	4.3	4.8	5.0	V
	Accuracy		-2	–	2	%
	Hysteresis	V _{SYS} falling	–	50	–	mV
V _{SYS_OV}	System over voltage protection threshold	Programmable (8 settings in 100 mV steps)	4.5	5.0	5.2	V
	Accuracy		-2	–	2	%
	Hysteresis	V _{SYS} falling	–	75	–	mV

1. On PM7250B chargers, the battery FET linear operation is removed, and precharge to fast-charge transition happens when VBAT reaches the V_{SYS_MIN} threshold.
2. The PM7250B charger can charge up to 4 A only.

3.5.3 Buck and boost regulator specifications

Table 3-8 Buck and boost regulator specifications

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
BOOT_PWR LDO						
$V_{\text{BOOT_PWR}}$	BOOT_PWR regulated output voltage	$I_{\text{OUT}} = 20 \text{ mA}$, $V_{\text{USB_IN}}/V_{\text{MID_CHG}} > 5.0 \text{ V}$	4.7	5	5.2	V
$V_{\text{BOOT_PWR_UVLO}}$	BOOT_PWR UVLO	$V_{\text{USB_IN}}/V_{\text{MID_CHG}}$ falling	–	3	–	V
PWM buck regulator						
$R_{\text{DSON_FP}}$	USB_IN input FET (12 V) on-resistance		–	12.1	20.1	mΩ
$R_{\text{DSON_HS}}$	High-side FET (20 V) on-resistance		–	16.5	27.6	mΩ
$R_{\text{DSON_LS}}$	Low-side FET (16 V) on-resistance		–	18.5	30.9	mΩ
$V_{\text{SYS_MIN}}$	Minimum VPH regulated output voltage range	Programmable (8 settings in 100 mV steps)	2.7	3.0	3.4	V
	Accuracy	500 mA load, $V_{\text{BAT}} < V_{\text{SYS_MIN}}$	-2	–	2	%
V_{SYS}	$V_{\text{PH_PWR}} (V_{\text{SYS}})$ output voltage	V_{IN} present, during trickle/precharging, $V_{\text{BAT}} < V_{\text{SYS_MIN}} - V_{\text{BAT_TRACK}}$	$V_{\text{SYS_MIN}}$			V
		V_{IN} present, during precharging, $V_{\text{SYS_MIN}} - V_{\text{BAT_TRACK}} < V_{\text{BAT}} < V_{\text{SYS_MIN}}$	$V_{\text{BAT}} + V_{\text{BAT_TRACK}}$			V
		V_{IN} present, during fast charging, $V_{\text{BAT}} > V_{\text{SYS_MIN}}$	$V_{\text{BAT}} + (I_{\text{CHG}} \times R_{\text{DSON_BF}})$			V
		V_{IN} present, charging terminated or disabled	$V_{\text{BAT}} + V_{\text{BAT_TRACK}}$			V
$V_{\text{BAT_TRACK}}$	V_{BAT} tracking range	Programmable (100 mV, 200 mV, 300 mV, 350 mV)	100	200	350	mV
	Accuracy		$\pm 10\%$ of setting $\pm 30 \text{ mV}$			–
$V_{\text{BOOT_UVLO}}$	Boot capacitor UVLO threshold	Boot UVLO enabled, 1-in-8 mode, $V_{\text{USB_IN}}/V_{\text{DC_IN}} = 5 \text{ V}$	2.2	3.2	4.2	V
$F_{\text{SW_BUCK}}$	Buck switching frequency range	Programmable	600	1050	1200	kHz
	Accuracy	$T = 0^\circ\text{C}$ to $+70^\circ\text{C}$	-6	–	6	%

Table 3-8 Buck and boost regulator specifications (cont.)

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
DC _{BUCK}	Buck duty cycle	Maximum, F _{sw} = 1.0 MHz, 1-in-8 mode	–	98.125	–	%
		Maximum, F _{sw} = 600 kHz, 1-in-8 mode	–	98.8	–	%
		Minimum	–	0	–	%
V _{DIP_BUCK}	Buck voltage dip due to positive load transient	V _{USB_IN} /V _{MID_CHG} = 5.0 V, I _{LIM_USB} = 3 A, V _{BATT} = 0 V, V _{PH} = V _{SYS_MIN} , F _{SW} = 600 kHz, I _{SYS_LOAD} ¹ 0.1 A to 2 A in 1 A/μs	-500	–	–	mV
V _{SPIKE_BUCK}	Buck voltage spike due to negative load transient	V _{USB_IN} /V _{MID_CHG} = 5/7/9/12 V, V _{BATT} = V _{FLT} (charging done) ¹ , F _{SW} , I _{SYS_LOAD} 4.0 A to 0.1 A in 1 A/μs	–	–	500	mV
V _{P-P_BUCK_PWM}	Buck output ripple in PWM mode	2 A load, V _{BATT} = 3.9 V, PWM mode, F _{SW} = 1.0 MHz ¹	–	25	50	mV
V _{P-P_BUCK_PFM}	Buck output ripple in PFM mode	No load, V _{BATT} = 3.0 V, PFM mode ¹	–	50	100	mV
I _{DD_BUCK}	Buck active supply current	Switcher enabled, no load, USB_IN = 5 V, V _{SYS} = V _{SYS_MIN} , PFM mode based on BOOT UVLO	–	2	10	mA
		Switcher enabled, no load, USB_IN = 5 V, V _{SYS} = 4.5 V, PWM mode	–	30	–	mA
PWM boost regulator						
V _{BOOST}	Boost regulated output voltage range	Programmable (8 settings in 100 mV steps)	4.8	5	5.5	V
	Boost regulated output voltage accuracy	All settings	-3	–	3	%
I _{BOOST}	Boost output current rating	V _{BATT} > V _{BAT_BST_UVLO}	1.5	–	–	A
V _{BAT_BST_UVLO}	Battery under-voltage lockout for low power boost (1.5 A)	Programmable (4 settings in 100 mV steps)	2.7	–	3.0	V
	Accuracy	V _{BATT} falling, all settings, T = 0°C to +70°C	-2	–	2	%
	Hysteresis	V _{BATT} rising, all settings, T=0°C to +70°C	25	50	75	mV

Table 3-8 Buck and boost regulator specifications (cont.)

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
I _{BOOST_OCP}	Steady-state boost output current limit range	Programmable (6 settings in 500 mA steps)	500	–	3000	mA
	Steady-state boost output current limit accuracy	VBATT = 4.0 V, FSW = 800 kHz, VBOOST = 5 V, IBOOST_LIM = 1.5 A	1.5	–	2.5	A
V _{BOOST_OCP}	Boost overcurrent shutdown threshold	I _{BOOST} > I _{BOOST_LIM} , V _{USB_IN} falling	–	1	–	V
V _{P-P_BOOST_PWM}	PWM boost output ripple	500 mA load, V _{BATT} = 4.0 V, PWM mode, F _{SW} = 800 kHz ¹	–	50	100	mV
V _{P-P_BOOST_PFM}	PFM boost output ripple	No load, V _{BATT} = 4.4 V, PFM mode ¹	–	100	150	mV
V _{DIP_BOOST}	Boost voltage dip due to positive load transient	VBATT = 4.0 V, FSW = 800 kHz, VBOOST = 5.0 V, IBOOST_LIM = 3.0 A, ILOAD 0.1 A to 1.5 A in 1 A/μs ¹	-500	–	–	mV
V _{SPIKE_BOOST}	Boost voltage spike due to negative load transient	VBATT = 4.0 V, FSW = 800 kHz, VBOOST = 5.0 V, IBOOST_LIM = 3.0 A, ILOAD 1.5 A to 0.1 A in 1 A/μs ¹	–	–	500	mV
I _{DD_BOOST}	Boost supply current	VBATT = 3.8 V, VBOOST = 5.0 V, PFM, OTG mode, no load	–	2	2.5	mA
		VBATT = 4.0 V, VBOOST = 5.0 V, PFM, Type-C source mode, Rp-3A pull-up (330 μA) enabled	–	2.5	3	mA
Eff_BOOST_PWM	Boost efficiency in PWM mode	VBATT = 3.8 V, PWM mode, FSW = 800 kHz, 1.5 A load	–	94	–	%
DC _{BOOST}	Boost duty cycle		0	–	75	%
Oscillator						
f _{OSC_STBY}	Standby oscillator frequency	For charger digital	–	200	–	kHz
	Accuracy	T = 25°C	-1	–	1	%
		T = 0°C to +70°C	-2	–	2	%

1. C_{sys} = 22 μF, C_{USB_IN} = 4.7 μF, C_{MID} = 4.7 μF

3.5.4 Battery interface specifications

Table 3-9 Battery interface specifications

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
Battery FET						
$R_{\text{DSON_BF}}$	BATT-to-SYS MOSFET on-resistance	Battery FET (5 V), ball-to-ball	–	6.0	10.1	mΩ
$I_{\text{BF_DC}}$	Maximum continuous DC battery FET current	Battery discharging	–	–	10	A
$I_{\text{BF_PEAK}}$	Peak battery FET current	Battery discharging, max 10 ms	–	–	12	A
Battery supplemental mode (BSM)						
$V_{\text{BSM_DET}}$	BSM detection comparator threshold	$V_{\text{PH_PWR}} - V_{\text{BAT_PWR}}$	-75	-50	-25	mV
$t_{\text{BSM_ENTRY}}$	BSM activation time	From $V_{\text{PH_PWR}}$ drop below $V_{\text{BSM_DET}}$ to battery FET starts to conduct	–	–	5	μs
$V_{\text{PH_DIP_BSM}}$	$V_{\text{PH_PWR}}$ voltage dip below $V_{\text{BAT_PWR}}$ upon BSM	$V_{\text{USB_IN}} = 5 \text{ V}$, $I_{\text{LIM_USB}} = 500 \text{ mA}$, $V_{\text{PH_PWR}} = 3.6 \text{ V}$, $V_{\text{BATT}} = 3.5 \text{ V}$, charging disabled; I_{SYS} rises from 100 mA to 3 A in 20 μs; $C_{\text{BAT}} = 10 \text{ μF}$, $C_{\text{SYS}} = 22 \text{ μF}$	–	–	200	mV
$I_{\text{BSM_TERM_ANA}}$	Analog BSM termination comparator range	Programmable (8 settings in 50 mA steps)	100	100	450	mA
	$I_{\text{BSM_TERM_ANA}}$ comparator accuracy	$T = 0^\circ\text{C}$ to $+70^\circ\text{C}$	$\pm 10\%$ of setting $\pm 50 \text{ mA}$			–
$t_{\text{BSM_EXIT_ANA}}$	Analog BSM termination time	From I_{BAT} across $I_{\text{BSM_TERM_ANA}}$ to battery FET stops conducting	–	–	5	μs

3.5.5 Miscellaneous specifications

Table 3-10 Miscellaneous specifications

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
Thermal control and protection						
$T_{\text{DIE_REG_L}}$	IC die temperature regulation window lower threshold	Programmable range in 1°C steps, T_{DIE} rising, from VADC digital comparator	60	–	130	$^\circ\text{C}$
		Accuracy	-3	–	3	$^\circ\text{C}$

Table 3-10 Miscellaneous specifications

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
T _{DIE_REG_H}	IC die temperature regulation window upper threshold	Programmable, TDIE_REG_L + X°C with 1°C, TDIE rising, VADC digital comparator	1	–	10	°C
		Accuracy	-3	–	3	°C
T _{SKIN_REG_L}	Skin temperature regulation window lower threshold	TSKIN ≥ TSKINREGL (programmable 10°C to 60°C in 0.5°C steps), skin temp rising, Qualcomm battery gauge digital comparator	-1	–	1	°C
T _{SKIN_REG_H}	Skin temperature regulation window upper threshold	TSKIN ≥ TSKINREGH (programmable, TSKINREGL + 1-10°C with 0.5°C step), skin temp rising, Qualcomm battery gauge digital comparator	-1	–	1	°C
T _{CONN_REG_L}	USB connector temperature regulation window lower threshold	Programmable range in 1°C steps	10	–	60	°C
		Accuracy	-1	–	1	°C
T _{CONN_REG_H}	USB connector temperature regulation window upper threshold		10	–	60	°C
		Accuracy	-1	–	1	°C
T _{TEMP_RST}	Rapid increase die temperature threshold	T _{DIE} rising, Programmable range in 10°C steps	100	120	130	°C
		Accuracy	-3	–	3	°C
		Hysteresis	3	–	5	°C
T _{SHDN}	IC thermal shutdown temp	T _{DIE} rising, Programmable range in 5°C steps	145	155	160	°C
		Accuracy	-3	–	3	°C
T _{SHDN_HYST}	IC thermal shutdown hysteresis	T _{DIE} falling	–	20	–	°C
Logic inputs/outputs						
V _{IL}	Input logic-low threshold	KPD_PWR_N	–	–	0.6	V
V _{IH}	Input logic-high threshold	KPD_PWR_N	1.4	–	–	V
VOL	Digital output low level	DC_IN_EN, CC_OUT, SMB_EN, SYS_OK, configured as CMOS push-pull output, or SYS_OK configured as NMOS open-drain output	–	0.1	0.3	V

Table 3-10 Miscellaneous specifications

Symbol	Parameter	Conditions	Minimum	Typical	Maximum	Units
VOH	Digital output high level	DC_IN_EN, CC_OUT, SMB_EN, SYS_OK, configured as CMOS push-pull output	1.5	1.8	–	V
R _{PULL}	Push-pull output pull-up resistance	Output configured as push-pull, VDD = 1.8 V	–	1.27	–	kΩ

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3.6 Qualcomm battery gauge

The Qualcomm battery gauge module offers a software-based algorithm that can accurately estimate the battery's state of charge by utilizing current monitoring and voltage-based techniques. This hybrid approach ensures both excellent short-term linearity and long-term accuracy. Furthermore, neither full battery charge cycling, nor zero-current-load conditions, are required to maintain the accuracy. This module utilizes dedicated ADCs to measure both battery current and battery voltage simultaneously. The battery current is measured through internal replica sensing of BATFET current. Same ADC measures the battery pack temperature by sensing the voltage at BAT_THERM pin. Using precise measurements of battery voltage, current, and temperature, the gauging algorithm compensates for the variation in battery characteristics across temperature changes and aging effects. This provides a dependable state of charge estimate throughout the entire life of the battery and across a broad range of operating conditions. Qualcomm battery gauge provides battery voltage and current for the software to read and run the gauge processing routines. All the supported Qualcomm battery gauge features are implemented by the software based on the readings supplied by the hardware.

Table 3-11 Qualcomm battery gauge specifications

Parameter	Comments	Minimum	Typical	Maximum	Units
Supported thermistor beta range		3300	–	4400	
Thermistor supported range		10	–	100	kΩ
BAT_ID resistor supported range		7.5	–	450	kΩ

Table 3-12 Qualcomm battery gauge performance specifications

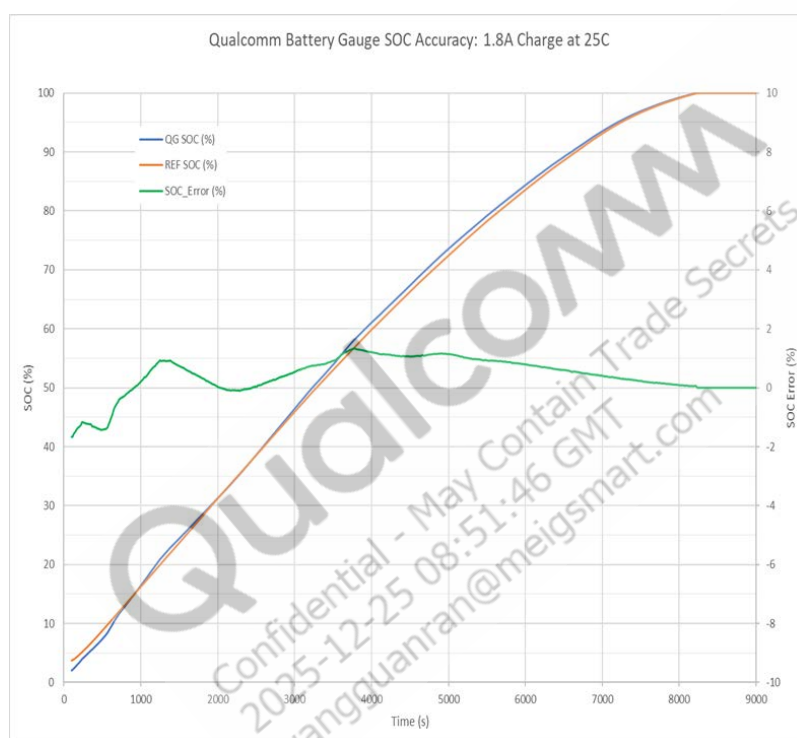
Specification	Test conditions	Min	Typ	Max	Unit
Battery temperature reading error using 100 K, 1%, 4250 K beta thermistor, and 100 K pull-up	Ratiometric calibration data (0°C to 50°C), VBAT = 2.7 to 5 V, IBAT = 0 A	-1.0	–	1.0	°C
	Ratiometric calibration data (-20°C to 70°C), VBAT = 2.7 to 5 V, IBAT = 0 A	-1.5	–	1.5	°C
Battery temperature reading error using 10 K to 100 K, 1%, 4250 K beta thermistor, and 30 K pull-up	Ratiometric calibration data (0°C to 50°C), VBAT = 2.7 to 5 V, IBAT = 0 A	-2.0	–	2.0	°C
	Ratiometric calibration data (-20°C to 70°C), VBAT = 2.7 to 5 V, IBAT = 0 A	-3.0	–	3.0	°C
Battery temperature reading error using 10 K to 100 K, 1%, 4250 K beta thermistor, and 100 K pull-up	Ratiometric calibration data (0°C to 50°C), VBAT = 2.7 to 5 V, IBAT = 0 A	-3.0	–	3.0	°C
	Ratiometric calibration data (-20°C to 70°C), VBAT = 2.7 to 5 V, IBAT = 0 A	-5.0	–	5.0	°C

Table 3-13 Battery interface module specifications

Parameter	Comments	Minimum	Typical	Maximum	Unit
Battery monitoring frequency and current drawn					
Active mode	Battery is monitored continuously	–	5	–	μA
Sleep mode	Battery is monitored every 1 ms	–	0.76	–	μA
POFF mode	Battery is monitored every 500 ms	–	0.0015	–	μA

Table 3-13 Battery interface module specifications (cont.)

Parameter	Comments	Minimum	Typical	Maximum	Unit
Supply voltage range for internal pull-up					
aVdd	With 1 MΩ internal pull-up resistance	–	1.8	2.2	V
Voltage detection threshold					
Range	Range for detection threshold as a percentage of aVdd supply voltage	36	–	96	%
Programmable step size		–	4	–	%
Accuracy	Detection threshold accuracy	-2	–	2	%

**Figure 3-1 Qualcomm battery gauge SoC accuracy plot at 1.8 A charge and 25°C**

Status: Charge; Current into battery – 1 A; Battery temperature: 25°C

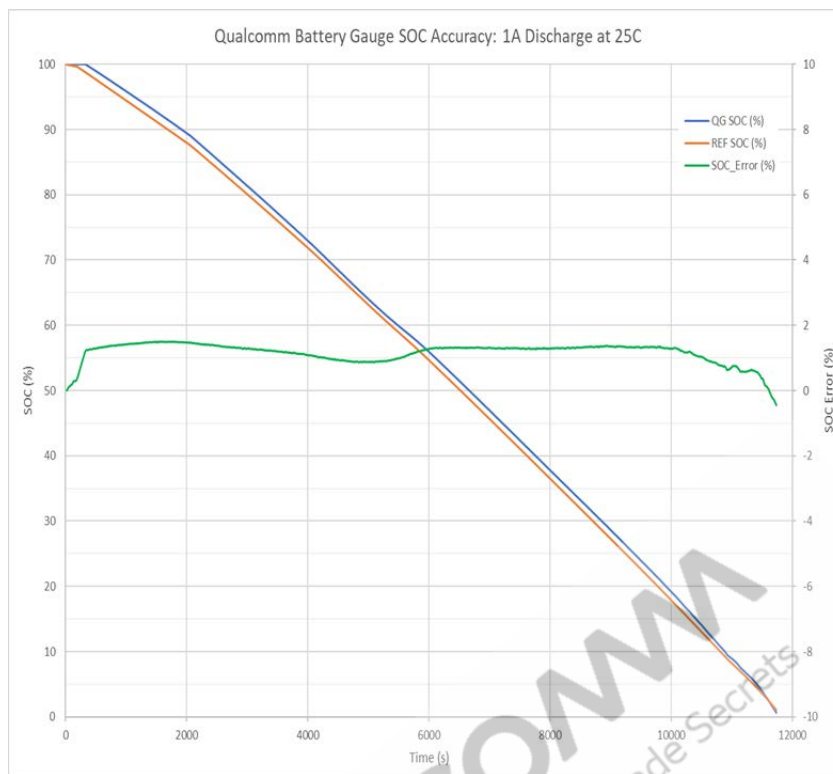


Figure 3-2 Qualcomm battery gauge SoC accuracy plot at 1 A discharge and 25°C

Status: Discharge; Current into battery – 1 A; Battery temperature: 25°C

3.7 Battery interface module

A dedicated module has been created for detecting the presence of the battery. This module standardizes the method and capability for capturing the battery removal. The battery presence is determined in the BMD hardware by using either the battery's internal BAT_ID resistor (default), or the battery's thermistor (BAT_THERM). One critical feature of the BIM module is to detect battery presence while drawing very low current from the battery. The low current drawn is extremely important in sleep/standby and off-mode detection, where each μA of current is critical. BIM also has the capability to deglitch the output value that helps in avoiding false detection when the phone is dropped, and the battery connection is open very briefly.

3.8 Battery current limiting (BCL) module

The BCL subsystem provides both hardware and software alarms, which are used by the system for reducing battery current to prevent the below mentioned two conditions:

- Battery over current protection (OCP)
- Battery under voltage protection (UVP)

The BCL module obtains input signals from the VADC, IADC, and two voltage comparators. It contains the appropriate set and clear deglitching capability for input signals and latched status registers for software to read and determine the alarm source. It has the capability to combine, encode, and generate interrupts. The alarm sources are combined into level 0, level 1, level 2 and BF (BAT_FET open) system alarms. The BCL module has the ability to route an alarm level to the BA_N output signal and the charger module.

Table 3-14 BCL specifications

Parameter	Comments	Minimum	Typical	Maximum	Unit
BCL: current sampling interval					
LPM	Since in LPM mode the system current is very low, the BCL sampling time interval can be larger	–	–	20	sec
HPM		–	–	1	ms
BCL: voltage sampling interval					
LPM		–	–	80	sec
HPM		–	–	100	ms
BCL: battery voltage comparator					
VBATT low comparator accuracy		-50	–	50	mV
VBATT too-low comparator accuracy		-50	–	50	mV

3.9 General housekeeping

The PMIC includes many circuits that support handset-level housekeeping functions - various tasks that must be performed to keep the handset in order. Integration of these functions reduces the external parts count and the associated size and cost. Housekeeping functions include an analog switch matrix, multiplexers, and voltage scaling; an ADC circuit and overtemperature protection.

3.9.1 Analog multiplexer and scaling circuits

A set of analog switches, analog multiplexers, and voltage-scaling circuits select and condition a single analog signal for routing to the on-chip ADC. The multiplexer and scaling functions are summarized in below table:

Table 3-15 ADC channels list

Channel number (HEX)	Description	Scaling	Internal pull-up	Source	Input range (V)
0	REF_GND	1/1	Open	Pin: REF_GND	0 to 1.875
1	1P25VREF	1/1	Open	Internal: MBG	0 to 1.875
2	VREF_VADC	1/1	Open	Internal: VADC LDO	0 to 1.875

Table 3-15 ADC channels list (cont.)

Channel number (HEX)	Description	Scaling	Internal pull-up	Source	Input range (V)
82	VREF_VADC	1/3	Open	Internal: VADC LDO	0 to 1.875
83	VPH_PWR	1/3	Open	VPH_PWR	0 to 5
84	VBAT_SNS	1/3	Open	VBATT_SNS_P	0 to 5
6	DIE_TEMP	1/1	Open	Internal: DIE_TEMP	0 to 1.875
8	USB_IN_V/16	1/16	Open	USB_IN	0 to 28
9	CHG_TEMP	1/1	Open	Internal: CHG_TEMP	0 to 1.875
A	BAT_THERM	1/1	Open	BAT_THERM	0 to 1.875
B	BAT_ID	1/1	Open	Pin: BATT_ID	0 to 1.875
C	OPTION	1/1	Open	Pin: OPTION	0 to 1.875
D	AMUX_THM1	1/1	Open	AMUX_1	0 to 1.875
E	AMUX_THM2	1/1	Open	SMB_THERM	0 to 1.875
F	AMUX_THM3	1/1	Open	CONN_THERM	0 to 1.875
10	AMUX_THM4	1/1	Open	GPIO1 (LV)	0 to 1.875
12	AMUX1_GPIO	1/1	Open	GPIO11 (LV)	0 to 1.875
13	AMUX2_GPIO	1/1	Open	GPIO12 (LV)	0 to 1.875
14	AMUX3_GPIO	1/1	Open	GPIO5 (MV)	0 to 1.875
15	AMUX4_GPIO	1/1	Open	GPIO8 (MV)	0 to 1.875
1E	MID_CHG/6	1/6	Open	MID_CHG	0 to 10.5
2A	BAT_THERM	1/1	30K	BAT_THERM	0 to 1.875
2B	BAT_ID	1/1	30K	BAT_ID	0 to 1.875
2C	OPTION	1/1	30K	OPTION	0 to 1.875
2D	AMUX_THM1	1/1	30K	AMUX_1	0 to 1.875
2E	AMUX_THM2	1/1	30K	SMB_THERM	0 to 1.875
2F	AMUX_THM3	1/1	30K	CONN_THERM	0 to 1.875
30	AMUX_THM4	1/1	30K	GPIO1 (LV)	0 to 1.875
32	AMUX1_GPIO	1/1	30K	GPIO11 (LV)	0 to 1.875
33	AMUX2_GPIO	1/1	30K	GPIO12 (LV)	0 to 1.875
34	AMUX3_GPIO	1/1	30K	GPIO5 (MV)	0 to 1.875
35	AMUX4_GPIO	1/1	30K	GPIO8 (MV)	0 to 1.875
4A	BAT_THERM	1/1	100K	BAT_THERM	0 to 1.875
4B	BAT_ID	1/1	100K	BAT_ID	0 to 1.875
4C	OPTION	1/1	100K	OPTION	0 to 1.875
4D	AMUX_THM1	1/1	100K	AMUX_1	0 to 1.875
4E	AMUX_THM2	1/1	100K	SMB_THERM	0 to 1.875
4F	AMUX_THM3	1/1	100K	CONN_THERM	0 to 1.875

Table 3-15 ADC channels list (cont.)

Channel number (HEX)	Description	Scaling	Internal pull-up	Source	Input range (V)
50	AMUX_THM4	1/1	100K	GPIO1 (LV)	0 to 1.875
52	AMUX1_GPIO	1/1	100K	GPIO11 (LV)	0 to 1.875
53	AMUX2_GPIO	1/1	100K	GPIO12 (LV)	0 to 1.875
54	AMUX3_GPIO	1/1	100K	GPIO5 (MV)	0 to 1.875
55	AMUX4_GPIO	1/1	100K	GPIO8 (MV)	0 to 1.875
6A	BAT_THERM	1/1	400K	BAT_THERM	0 to 1.875
6B	BAT_ID	1/1	400K	BAT_ID	0 to 1.875
6C	OPTION	1/1	400K	OPTION	0 to 1.875
6D	AMUX_THM1	1/1	400K	AMUX_1	0 to 1.875
6E	AMUX_THM2	1/1	400K	SMB_THERM	0 to 1.875
6F	AMUX_THM3	1/1	400K	CONN_THERM	0 to 1.875
70	AMUX_THM4	1/1	400K	GPIO1 (LV)	0 to 1.875
72	AMUX1_GPIO	1/1	400K	GPIO11 (LV)	0 to 1.875
73	AMUX2_GPIO	1/1	400K	GPIO12 (LV)	0 to 1.875
74	AMUX3_GPIO	1/1	400K	GPIO5 (MV)	0 to 1.875
75	AMUX4_GPIO	1/1	400K	GPIO8 (MV)	0 to 1.875
94	AMUX3_GPIO	1/3 ¹	Open	GPIO5 (MV)	0 to 5
95	AMUX4_GPIO	1/3 ¹	Open	GPIO8 (MV)	0 to 5

1. Voltage on MV GPIOs should not exceed instantaneous voltage on VPH_PWR.

3.9.2 ADC specifications

The analog to digital converter circuit (VADC and IADC) of PM7250B is shared by the battery thermal monitor (BTM), Qualcomm battery gauge, BCL, charger, user, and other housekeeping functions.

Table 3-16 ADC performance specifications

Parameter	Comment	Minimum	Typical	Maximum	Unit
Voltage conversion (VADC)					
1/1 channel end-to-end accuracy ¹	T _{amb} = -20°C to 70°C, VBATT = 2.5 V to 4.75 V	-11	±6	11	mV
1/1 channel end-to-end accuracy with internal pull-up ²	T _{amb} = -20°C to 70°C, VBATT = 2.5 V to 4.75 V	-12.5	±7	12.5	mV
1/3 channel end-to-end accuracy ³	T _{amb} = -20°C to 70°C, VBATT = 2.5 V to 4.75 V	-20	±10	20	mV
ADC resolution (LSB)	1/1 channel		64.879		μV
	Scaled to 1/3 channel		194.637		μV

Table 3-16 ADC performance specifications (cont.)

Parameter	Comment	Minimum	Typical	Maximum	Unit
Guaranteed input battery voltage range		2.5	–	4.75	V
100 K pull-up	Trimmed value	99.5	100	100.5	kΩ
400 K pull-up	Trimmed value	398	400	402	kΩ
30 K pull-up	Trimmed value	29.7	30	30.3	kΩ
Internal pull-up accuracy	100 K and 400 K pull-up	–	0.5	–	%
	30 K pull-up	–	1	–	%
Pull-up temperature coefficient	–	-200	–	200	ppm/°C
1/1 channel AMUX input resistance	–	10	–	–	MΩ
1/3 channel AMUX input resistance	–	1	–	–	MΩ
1/3 channel end-to-end accuracy on VBAT_SNS to PACK_SNS inputs	T _{amb} = 25°C, VBATT = 2.5 V to 5.25 V	-0.2	–	0.2	%
1/3 channel end-to-end accuracy on VBAT_SNS to PACK_SNS inputs	T _{amb} = -20°C to 70°C, VBATT = 2.5 V to 5.25 V	-0.3	–	0.3	%
Current conversion (IADC with internal sensing)					
Battery current sensing range	VBATT = 3.0 V to 4.75 V T _{amb} = 0°C to 70°C	-10	–	10	A
Battery current sensing offset	VBATT = 3.0 V to 4.75 V IBAT = 0 to 8.5 A T _{amb} = 25°C	-11	–	11	mA
Battery current sensing offset	VBATT = 3.0 V to 4.75 V IBAT = 0 to 8.5 A T _{amb} = 0°C to 70°C	-15	–	15	mA
Battery current gain error	VBATT = 3.0 V to 4.75 V IBAT = 0 to 1 A T _{amb} = 0°C to 70°C	-2	–	2	%
Battery current gain error	VBATT = 3.0 V to 4.75 V IBAT = 1 A to 5 A T _{amb} = 25 °C	-3	–	3	%
Battery current gain error	VBATT = 3.0 V to 4.75 V IBAT = 1 A to 5 A T _{amb} = 0°C to 70°C	-4	–	4	%
Battery current gain error	VBATT = 3.0 V to 4.75 V IBAT = 5 A to 8.5 A T _{amb} = 25 °C	-3.5	–	3.5	%

Table 3-16 ADC performance specifications (cont.)

Parameter	Comment	Minimum	Typical	Maximum	Unit
Battery current gain error	VBATT = 3.0 V to 4.75 V IBAT = 5 A to 8.5 A T _{amb} = 0°C to 70°C	-5	–	5	%
ADC resolution LSB	–	–	305.185	–	μA
Current conversion (IADC with external sensing resistor)					
Battery current sensing range	VBATT = 3.0 V to 4.75 V T _{amb} = 0°C to 70°C	-10	–	10	A
Battery current sensing offset	VBATT = 3.0 V to 4.75 V IBAT = 0 to 10 A T _{amb} = 25°C	-6	–	6	mA
Battery current sensing offset	VBATT = 3.0 V to 4.75 V IBAT = 0 to 10 A T _{amb} = 0°C to 70°C	-6	–	6	mA
Battery current relative error	VBATT = 3.0 V to 4.75 V IBAT = 0 to 10 A T _{amb} = 25°C	-0.75	–	0.75	%
Battery current relative error	VBATT = 3.0 V to 4.75 V IBAT = 0 A to 10 A T _{amb} = 0°C to 70°C	-1	–	1	%
ADC resolution LSB	–	–	305.185	–	μA
Sense resistor value	Sensing range programmable	2	5	10	mΩ
Sense resistor accuracy	–	–	0.5	1	%
Current sensing noise	VBATT = 3.0 V to 4.75 V IBAT = 1 A to 10 A T _{amb} = 0°C to 70°C	–	5	–	mA P-P
Common parameters					
Analog bandwidth (anti-alias filter)	–	–	500	–	kHz
ADC LDO voltage	–	1.828	1.875	1.910	V
ADC sample clock	–	–	4.8	–	MHz
ADC conversion time	1 K decimation ratio, 4.8 MHz sample clock	–	654	700	μs
Current consumption - ADC only	VADC active - does not include other PMIC infrastructure	–	450	500	μA
Current consumption - rock bottom sleep current	PMi in sleep, current consumed by ADC in RBS mode	–	5	10	μA

1. This 1/1 channel end-end accuracy applies to all measurements that are selected with 1/1 scaling without internal pull-up and resolution is 64.879 μV.
2. This 1/1 channel end-end accuracy applies to all measurements that are selected with 1/1 scaling with internal pull-up. For example, BAT_THERM, BAT_ID, and so on and resolution is 64.879 μV.

3. This 1/3 channel end-end accuracy applies to all measurements that are selected with 1/3 scaling. For example, BATT_PWR, VPH_PWR, and so on and resolution is 194.637 μ V

3.10 VIB_DRV module

VIB_DRV uses vibration to communicate an event or action through human touch. In a mobile phone, VIB_DRV is used to simulate the feeling of a real mechanical key by providing tactile feedback to the user as confirmation of touchscreen contact, or dynamic feedback to enhance the user's gaming experience. Pertinent performance specifications are listed in [Table 3-17](#).

Table 3-17 VIB_DRV performance specifications

Parameter	Comments	Minimum	Typical	Maximum	Units
Input voltage range	–	2.5	–	5.5	V
Output voltage range	–	1.504	–	3.544	V
V _{OUT} step size	–	–	8	–	mV
Rated load current	Higher load current is possible at increased headroom	–	300	–	mA
Overall DC error	Measured at Cout (output capacitor) $I_RATIO = I_LOAD/I_RATED$	- 5% - (6 mV+15 mV) $\times I_RATIO$	–	5	%Vout
Startup settling time	To within 1% of final value	130	200	550	μ s
Startup in-rush current	During start-up. variation due to bulk cap size.	–	25	300	mA
Overcurrent protection detection threshold	Threshold for OCP detection and no variable current limit settings	450	650	700	mA

3.11 GPIO specifications

The 10 GPIO ports are digital I/Os that can be programmed for a variety of configurations (Table 3-18). Performance specifications for the different configurations are included in Section 3.4.

NOTE: All unused GPIO pins, except for the following, should be configured as digital inputs with 10 μ A pull-down (their default state).

Table 3-18 Programmable GPIO configurations

Configuration type	Configuration description
Input	- No pull-up - Pull-up (1.5 μA, 30 μA, or 31.5 μA) - Pull-down (10 μA)
Output	Open-drain or CMOS Inverted or noninverted Programmable drive current

Most GPIOs default to digital input with 10 μ A pull-down at power-on. Before they can be used for their intended purposes, they must be configured for use.

GPIOs are designed to run at a 4 MHz rate to support high-speed applications. The supported rate depends on the load capacitance and IR drop requirements. If the application specifies load capacitance, then the maximum rate is determined by the IR drop. If the application does not require a specific IR drop, then the maximum rate can be increased by increasing the supply voltage and adjusting the drive strength according to the actual load capacitance.

Table 3-19 GPIO_MV digital I/O characteristics

Parameter		Comments ¹	Min	Typ	Max	Units
V_{IH}	High-level input voltage	—	$0.65 \times V_{IO}$	—	—	V
V_{IL}	Low-level input voltage	—	—	—	$0.35 \times V_{IO}$	V
V_{SHYS}	Schmitt hysteresis voltage	—	$0.1 \times V_{IO}$	—	—	V
I_L	Input leakage current ²	—	-0.20	—	+0.20	μ A
V_{OH}	High-level output voltage	—	$0.8 \times V_{IO}$	—	V_{IO}	V
V_{OL}	Low-level output voltage	—	0	—	$0.2 \times V_{IO}$	V

1. V_{IO} is the supply voltage for the SoC/PMIC interface (most PMIC digital I/Os).

2. GPIO pins comply with the input leakage specification only when configured as a digital input or set to the tri-state mode.

Table 3-20 GPIO_MV VOL and VOH and minimum load current

Supply voltage	VOL, VOH	Minimum load current		
		Low strength driver	Medium strength driver	High strength driver
1.8 V	VOH = $0.8 \times V_{IO}$	0.15 mA	0.3 mA	0.6 mA
	VOL = $0.2 \times V_{IO}$			
3.6 V	VOH = $0.8 \times V_{IO}$	0.25 mA	0.5 mA	1 mA
	VOL = $0.2 \times V_{IO}$			

Table 3-21 GPIO_LV digital I/O characteristics

Parameter		Comments ¹	Min	Typ	Max	Units
V _{IH}	High-level input voltage	–	0.9	–	–	V
V _{IL}	Low-level input voltage	–	–	–	0.6	V
V _{SHYS}	Schmitt hysteresis voltage	–	180	–	–	mV
I _L	Input leakage current ²	–	-0.20	–	+0.20	μA
V _{OH}	High-level output voltage	–	$0.8 \times V_{IO}$	–	V _{IO}	V
V _{OL}	Low-level output voltage	–	0	–	$0.2 \times V_{IO}$	V

1. V_{IO} is the supply voltage for the SoC/PMIC interface (most PMIC digital I/Os).

2. GPIO pins comply with the input leakage specification only when configured as a digital input or set to the tri-state mode.

Table 3-22 GPIO_LV VOL and VOH and minimum load current

Supply voltage	VOL, VOH	Minimum load current		
		Low strength driver	Medium strength driver	High strength driver
1.8 V	VOH = $0.8 \times V_{IO}$	1 mA	1.5 mA	2 mA
	VOL = $0.2 \times V_{IO}$			

4 Mechanical information

4.1 Device physical dimensions

The PM7250B is available in the FOWPSP110 package that includes ground pins for improved grounding, mechanical strength, and thermal continuity. The FOWPSP110 package has a 3.51 mm × 4.40 mm body with a maximum height of 0.55 mm. Pad 1 is located by an indicator mark on the top of the package. A simplified version of the FOWPSP110 package outline drawing is shown in [Figure 4-1](#).

NOTE: Click the following link to download *Package Outline Drawing, FOWPSP110, 3.51 × 4.40 × 0.55 mm, M328* (NT90-PL897-1) from the Qualcomm® CreatePoint website.

<https://docs.qualcomm.com/bundle/NT90-PL897-1/resource/NT90-PL897-1.pdf>

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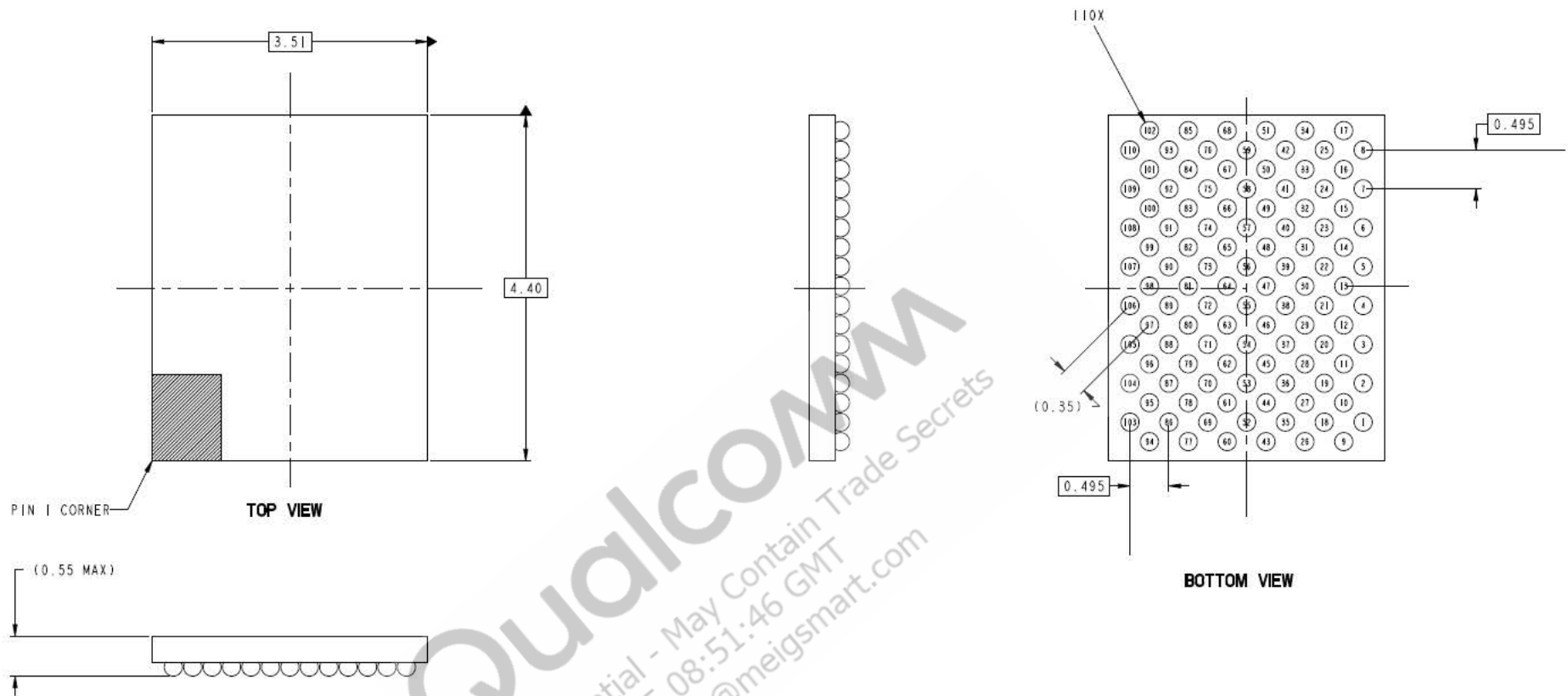


Figure 4-1 FOWPSP110 package outline drawing

NOTE: This is a simplified outline drawing. Click the link on the previous page to download the complete, up-to-date package outline drawing.

4.2 Part marking

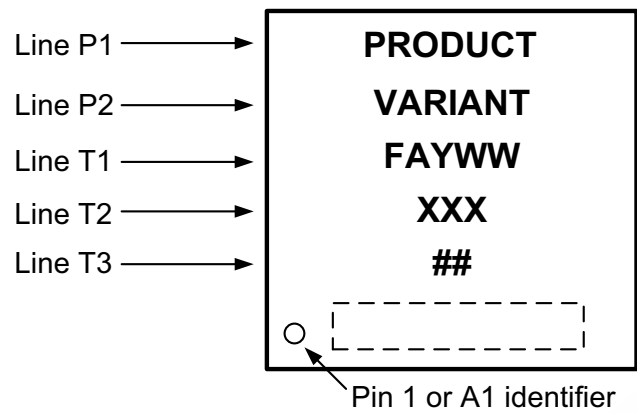


Figure 4-2 PM7250B device marking (top view, not to scale)

Table 4-1 PM7250B device marking line definitions

Line	Marking	Description
P1	PRODUCT	Qualcomm Technologies, Inc. (QTI) product name ■ PM7250B
P2	VARIANT	Device variant information ■ See Table 4-3 for the assigned values.
T1	FAYWW	F = supply source code ■ F = P for SMIC ■ F = J for Samsung A = assembly site (ball drop) code ■ A = E for ASE, Taiwan ■ A = T for DECA, Philippines Y = single/last digit of year WW = two-digit work week of year specified by Y
T2	XXX	XXX = traceability number
T3	##	## = wafer number
	○	Pin 1 or pin A1 indicator
	Blank or random	Additional content, as necessary

4.3 Device ordering information

4.3.1 Specification-compliant devices

The Oracle short description is used to order QTI products, and is present on both the customer label and this document. The short description includes the product name, configuration code, package type, product revision code, source code, and feature code/program ID of the part.

This device can be ordered using the identification code shown in [Figure 4-2](#).

Table 4-2 Device identification code

Device ID code	AAA-AAAA	-P	-TTTTTT	NNNN	A	+FF	-EE	-RR	-S	-BB or -PID ¹
Symbol definition	Product name	Configuration code	Package type	Number of pins	Package variable	Additional package information	Shipping package	Product revision	Source code	Feature code
Example 1	PM-7250B	-0	-FOWPSP	110			-TR	-00	-0	
Example 1	PM-7250B	-2	-FOWPSP	110			-TR	-00	-1	

1. The feature code (BB) and the program ID (PID) are mutually exclusive. A product may have one of them or none of them, but it will never have both. If there is no feature code/program ID, this field is blank, and the Oracle short description ends after the source configuration code (S).

For example:

- Example 1: PM-7250B-0-FOWPSP110-TR-00-0
- Example 2: PM-7250B-2-FOWPSP110-TR-00-1

Device identification details for all samples available to date are summarized in [Table 4-3](#).

Table 4-3 Device identification details

Device	Sample type	Product configuration code (P)	Product revision code (RR)	Feature code (BB) ¹	Hardware version	Source configuration code (S) ²
PM7250B	ES/CS ³ CS date codes are as follows: SMIC: ASE: 937 DECA: 941	0	00	—	v1.0	0
PM7250B-2	ES/CS ³ CS date codes are as follows: SMIC: ASE:117 DECA: 126	2	00	—	v1.0	0
PM7250B-2	CS ³ CS date codes are as follows: SMIC, Samsung: ASE:446 DECA: 446	2	00	—	v1.0	1

1. BB is the feature code that identifies an IC's specific feature set, which distinguishes it from other versions or variants.
2. S is the source configuration code that identifies all the qualified die fabrication-source combinations available when the particular sample type was shipped. S values are defined in [Table 4-4](#).
3. PM7250B ES and CS parts have the same PRR code.
All devices with YWW date code or later are CS quality. However, devices with earlier date codes may also be CS quality. Contact your customer service representative for CS part marking with date codes prior to YWW.

Table 4-4 Source configuration code

S value	F value = P	F value = J	A value = E	A value = T
0	SMIC	-	ASE, Taiwan	DECA, Philippines
1	SMIC	Samsung	ASE, Taiwan	DECA, Philippines
Other columns and rows will be added in future revisions of this document, if needed				

4.4 Thermal characteristics

Rather than provide thermal resistance values θ_{JC} and θ_{JA} , validated thermal package models are provided through the CreatePoint website. Designers can extract thermal resistance values by conducting their own thermal simulations.

NOTE: Click the following link to download the *PM7250B 110FOWPSP Package Thermal Model Icepak* (HS11-PM008-5HW) from the CreatePoint website.

<https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/HS11-PM008-5HW>

Click the following link to download the *PM7250B 110FOWPSP Package Thermal Model Flowtherm* (HS11-PM008-6HW) from the CreatePoint website.

<https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/HS11-PM008-6HW>

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5 Carrier, storage, and handling information

5.1 Carrier

5.1.1 Tape and reel information

All QTI carrier tape systems conform to EIA-481 standards.

A simplified sketch of the PM7250B tape carrier is shown in [Figure 5-1](#), including the proper part orientation, maximum number of devices per reel, and key dimensions.

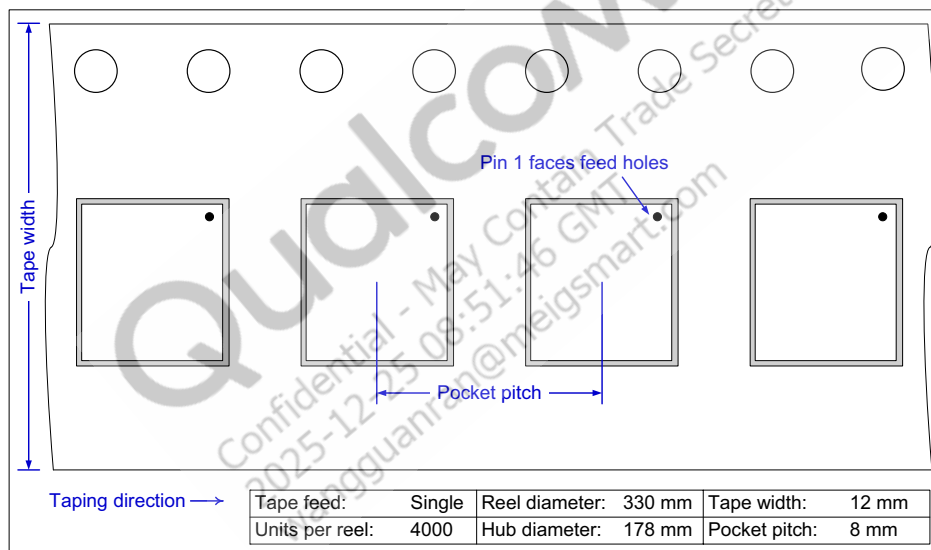


Figure 5-1 Carrier tape drawing with part orientation

Tape-handling recommendations are shown in [Figure 5-2](#).

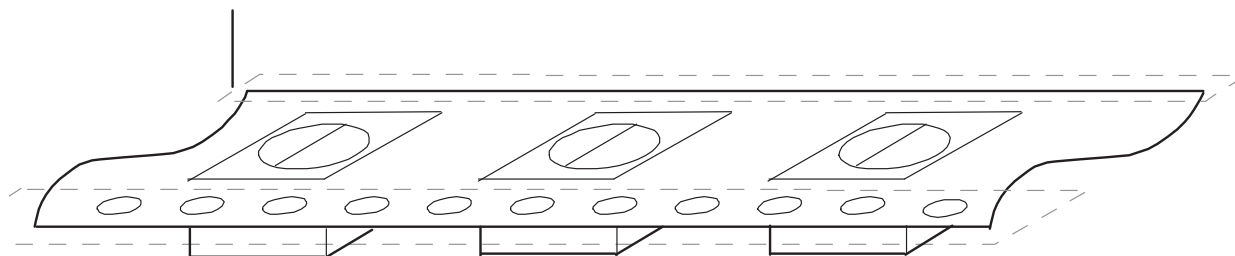


Figure 5-2 Tape handling

5.2 Device moisture-sensitivity level

A package's moisture sensitivity level (MSL) indicates its ability to withstand exposure after it is removed from its shipment bag, while it is on the factory floor awaiting PCB installation. A low MSL rating is better than a high rating; a low MSL device can be exposed on the factory floor longer than a high MSL device. All pertinent MSL ratings are summarized in [Table 5-1](#).

Table 5-1 MSL ratings summary

MSL	Out-of-bag floor life	Comments
1	Unlimited	$\leq 30^{\circ}\text{C}/85\% \text{ RH}$
2	1 year	$\leq 30^{\circ}\text{C}/60\% \text{ RH}$
2a	4 weeks	$\leq 30^{\circ}\text{C}/60\% \text{ RH}$
3	168 hrs.	$\leq 30^{\circ}\text{C}/60\% \text{ RH}$; PM7250B rating
4	72 hrs.	$\leq 30^{\circ}\text{C}/60\% \text{ RH}$
5	48 hrs.	$\leq 30^{\circ}\text{C}/60\% \text{ RH}$
5a	24 hrs.	$\leq 30^{\circ}\text{C}/60\% \text{ RH}$
6	Mandatory bake before use. After bake, must be reflowed within the time limit specified on the label.	$\leq 30^{\circ}\text{C}/60\% \text{ RH}$

QTI follows the latest IPC/JEDEC J-STD-020 standard revision for moisture-sensitivity qualification. *The PM7250B devices are classified as MSL3; the qualification temperature was $260^{\circ}\text{C} + 0^{\circ}/-5^{\circ}\text{C}$.* This qualification temperature ($260^{\circ}\text{C} + 0^{\circ}/-5^{\circ}\text{C}$) should not be confused with the peak temperature within the recommended solder reflow profile.

5.3 Storage

5.3.1 Bagged storage conditions

PM7250B devices delivered in tape and reel carriers must be stored in sealed, moisture barrier, antistatic bags. See the *IC Products Packing Method* (80-VK055-1) for the expected shelf life.

5.3.2 Out-of-bag duration

The out-of-bag duration is the time a device can be on the factory floor before being installed onto a PCB. It is defined by the device MSL rating, as described in [Section 4.4](#).

5.4 Handling

Tape handling was described in [Section 5.1.1](#). Other (IC-specific) handling guidelines are presented below.

5.4.1 Baking

Wafer-level packages such as the FOWPSP110 should not be baked.

5.4.2 Electrostatic discharge

Electrostatic discharge (ESD) occurs naturally in laboratory and factory environments. An established high-voltage potential is always at risk of discharging to a lower potential. If this discharge path is through a semiconductor device, destructive damage may result.

ESD countermeasures and handling methods must be developed and used to control the factory environment at each manufacturing site.

QTI products must be handled according to the ESD Association standard: ANSI/ESD S20.20-1999, *Protection of Electrical and Electronic Parts, Assemblies, and Equipment*.

5.5 Bar code label and packing for shipment

See the *IC Products Packing Method* (80-VK055-1) for all packing-related information, including bar code label details.

6 PCB mounting guidelines

6.1 RoHS compliance

The device complies with the requirements of the EU RoHS directive. Its SnAgCu solder balls use SAC405 composition. A product material declaration (PMD) that provides RoHS and other product environmental governance information is published when the data is available.

6.2 SMT assembly guidelines

For recommendations on SMT process development, see the *SMT Assembly Guidelines* (SM80-P0982-1).

NOTE: Click the following link to download the *SMT Assembly Guidelines* (SM80-P0982-1) from the CreatePoint website.

<https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/SM80-P0982-1>

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7 Part reliability

7.1 Reliability qualifications summary

Table 7-1 Silicon reliability results: SMIC/Samsung

Device qualification Tests, standards, and conditions	Sample size (SMIC/Samsung) number of lots	Result
Early life failure rate (ELFR) in DPPM HTOL: JESD22-A108-A Total samples from three different wafer lots	> 1000	Pass DPPM < 1000
Average failure rate (AFR) in FIT (λ) Failures per billion device hours; functional HTOL: JESD22-A108-A (Total samples from three different wafer lots)	231	Pass FIT < 50
Mean time to failure (MTTF) $t = 1/\lambda$ in million hours Total samples from three different wafer lots	231	Pass > 20
ESD – human body model (HBM) rating JESD22-A114-F	3	Pass
ESD – charged device model (CDM) rating JESD22-C101-E	3	Pass
Latch-up (I-test): EIA/JESD78A Trigger current: ± 100 mA; temperature: 85°C	3	Pass
Latch-up (Vsupply overvoltage): EIA/JESD78A Trigger voltage: each VDD pin, stress at $1.5 \times V_{ddmax}$ per device specification; temperature: 85°C	3	Pass

Table 7-2 Package reliability results for SMIC/Samsung

Package qualification Tests, standards, and conditions	Assembly source sample size (ASE)	Assembly source sample size (DECA)	Result
Moisture resistance test (MRT): MSL3; J-STD-020- D 3 × reflow cycles at 255°C +5/-0°C	693 231 × 3 assembly lots	693 231 × 3 assembly lots	Pass
Temperature cycle: JESD22-A104-D Temperature: -55°C to 125°C; number of cycles: 1000 Minimum soak time at minimum/maximum temperature: 5 minutes Cycle rate: 2 cycles per hour (CPH) MSL3 preconditioning: JESD22-A113-F Reflow temperature: 255°C +5/-0°C	231 77 × 3 assembly lots	231 77 × 3 assembly lots	Pass
Unbiased highly accelerated stress test (UHAST): JESD22-A118 MSL3 preconditioning: JESD22-A113-F Reflow temperature: 255°C +5/-0°C	231 77 × 3 assembly lots	231 77 × 3 assembly lots	Pass
Temperature cycle: JESD22-A104E Temperature: -55°C to 125°C; Soak time at minimum/maximum temperature: 8–10 minutes Cycle rate: 2 cycles per hour (CPH), 1000 cycles Preconditioning: JESD22-A113G MSL 3, reflow temperature: 260 +0/-5°C (Total samples from three different assembly lots)	231 77 × 3 assembly lots	231 77 × 3 assembly lots	Pass
High-temperature storage life: JESD22-A103-D Temperature 150°C, 1000 hours	231 77 × 3 assembly lots	231 77 × 3 assembly lots	Pass
Physical dimensions: JESD22-B100-A	15 15 × 1 assembly lots	15 15 × 1 assembly lots	Pass
Solder ball shear: JESD22-B117-B Total samples from three different assembly lots at each SAT	45 15 × 3 assembly lots	45 15 × 3 assembly lots	Pass

7.2 Qualification sample description

Table 7-3 Device characteristics

Category	Definition
Device name	PM7250B/PM7250B-2
Package type	FOWPSP110
Package body size	3.51 mm × 4.40 mm × 0.54 mm
Solder ball composition	SAC405
Process	150 nm
Fab sites	■ SMIC ■ Samsung
Assembly sites	■ ASE, Taiwan ■ DECA, Philippines
Solder ball pitch	0.35 mm

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8 Revision history

Bars appearing in the margin (as shown here) indicate where technical changes have occurred for this revision. The following table lists the technical content changes for all revisions.

Revision	Date	Description
A	March 2019	Initial release
B	January 2020	■ Table 2-2 <i>Pad descriptions – input power management functions</i>: Updated the pad type for pins 54 and 63 ■ Chapter 3 <i>Electrical specifications</i>: Added this chapter ■ Section 4.2 <i>Part marking</i>: Added this section ■ Section 4.3 <i>Device ordering information</i>: Added this section ■ Section 4.4 <i>Device moisture-sensitivity level</i>: Added this section ■ Section 4.5 <i>Thermal characteristics</i>: Added this section ■ Chapter 5 <i>Carrier, storage, and handling information</i>: Added this chapter ■ Chapter 6 <i>PCB mounting guidelines</i>: Added this chapter ■ Chapter 7 <i>Part reliability</i>: Added this section
C	April 2021	■ Table 3-12 <i>Qualcomm battery gauge performance specifications</i>: Added this table ■ Table 3-15 <i>ADC channels list</i>: Updated this table ■ Table 3-16 <i>ADC performance specifications</i>: Updated this table
D	July 2021	■ Section 1.3 <i>PM7250B SKUs</i>: Added this section ■ Table 4-2 <i>Device identification code</i>: Updated this table ■ Table 4-3 <i>Device identification details</i>: Updated this table
E	July 2022	■ Table 1-2 <i>PM7250B SKUs</i>: Updated the chipset attaches for PM7250B and PM7250B-2 SKUs ■ Table 4-3 <i>Device identification details</i>: Updated the footnote
F	April 3, 2023	■ Table 1-2 <i>PM7250B SKUs</i>: Updated the PM7250B-2 SKU list ■ Table 4-3 <i>Device identification details</i>: Updated the footnote
G	April 20, 2023	■ Table 1-2 <i>PM7250B SKUs</i>: Updated the PM7250B-2 SKU list ■ Table 3-3 <i>DC power supply currents</i>: Updated the parameter values

Revision	Date	Description
H	August 2023	■ Table 1-1 <i>PM7250B features</i>: Updated the peak efficiency for battery charger ■ Table 2-4 <i>Pad descriptions – General purpose input/output functions</i>: Updated the configurable function for GPIO_7 ■ Added the following tables: □ Table 3-19 <i>GPIO_MV digital I/O characteristics</i> □ Table 3-20 <i>GPIO_MV VOL and VOH and minimum load current</i> □ Table 3-21 <i>GPIO_LV digital I/O characteristics</i> □ Table 3-22 <i>GPIO_LV VOL and VOH and minimum load current</i>
Revision I was omitted in accordance with QTI document conventions.		
J	March 2025	■ Section Key features (see Section 1.2 for details): Updated supported interfaces to SPMI ■ Table 1-1 PM7250B features: Updated supported interfaces to SPMI ■ Table 2-4 Pad descriptions – General purpose input/output functions: Updated the configurable function for GPIO_1 to spare ■ Table 2-7 Pad descriptions – common grounds: Updated the pad # from 80 to 89 ■ Table 4-1 PM7250B device marking line definitions: Added Samsung to the foundry list ■ Table 4-2 Device identification code: Updated the source code for PM7250B to 1 ■ Table 4-3 Device identification details: Added CS details for PM7250B-2 ■ Table 4-4 Source configuration code: Added details for Samsung foundry ■ Table 7-1 Silicon reliability results: SMIC/Samsung: Updated the sample size details ■ Table 7-2 Package reliability results for SMIC/Samsung: Updated the sample size details ■ Table 7-3 Device characteristics: □ Updated the device name □ Added Samsung to the foundry list

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